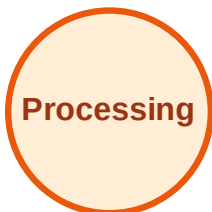


# BFS Traversal

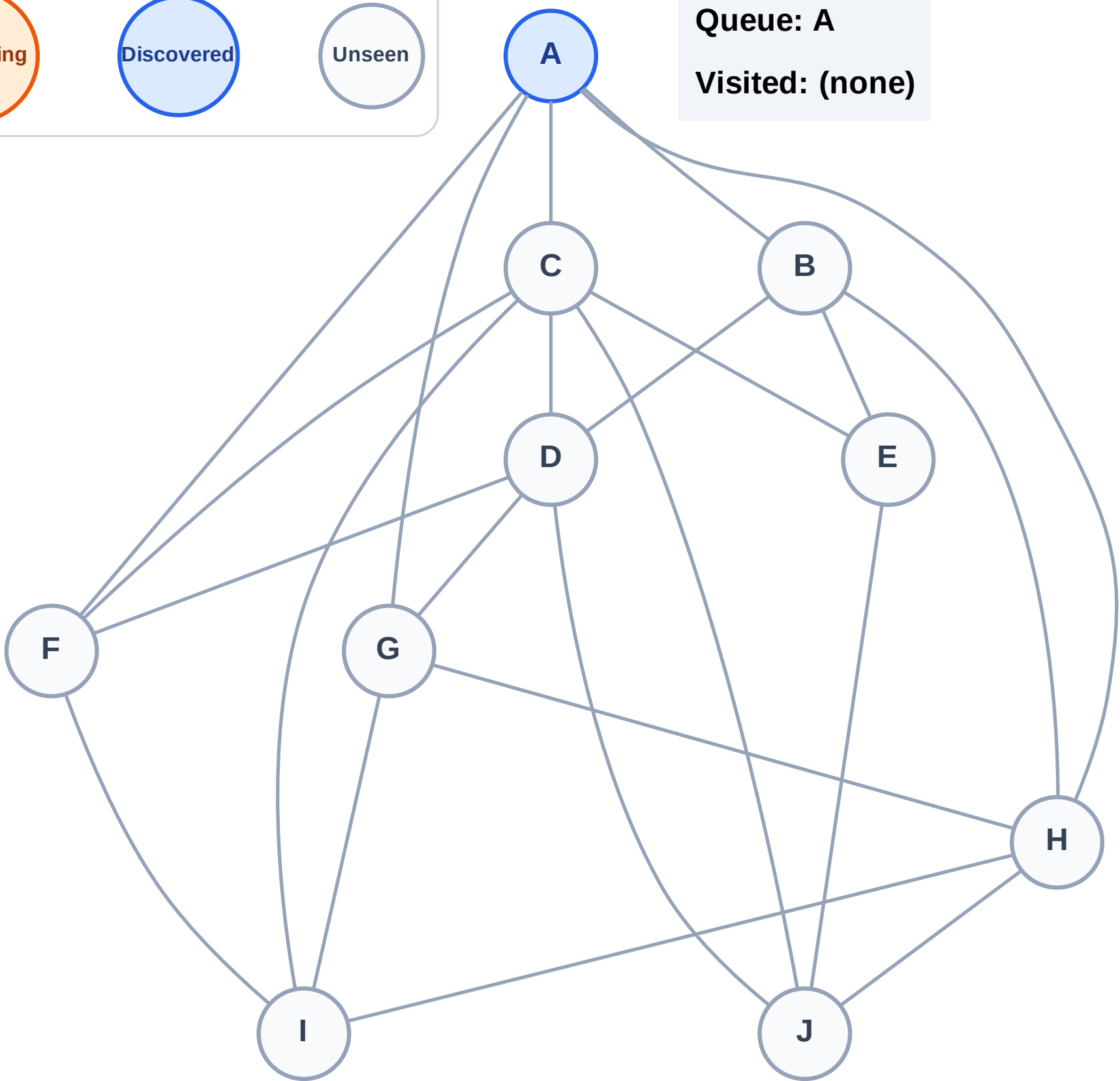
Step 1: Start at A

## Legend



Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

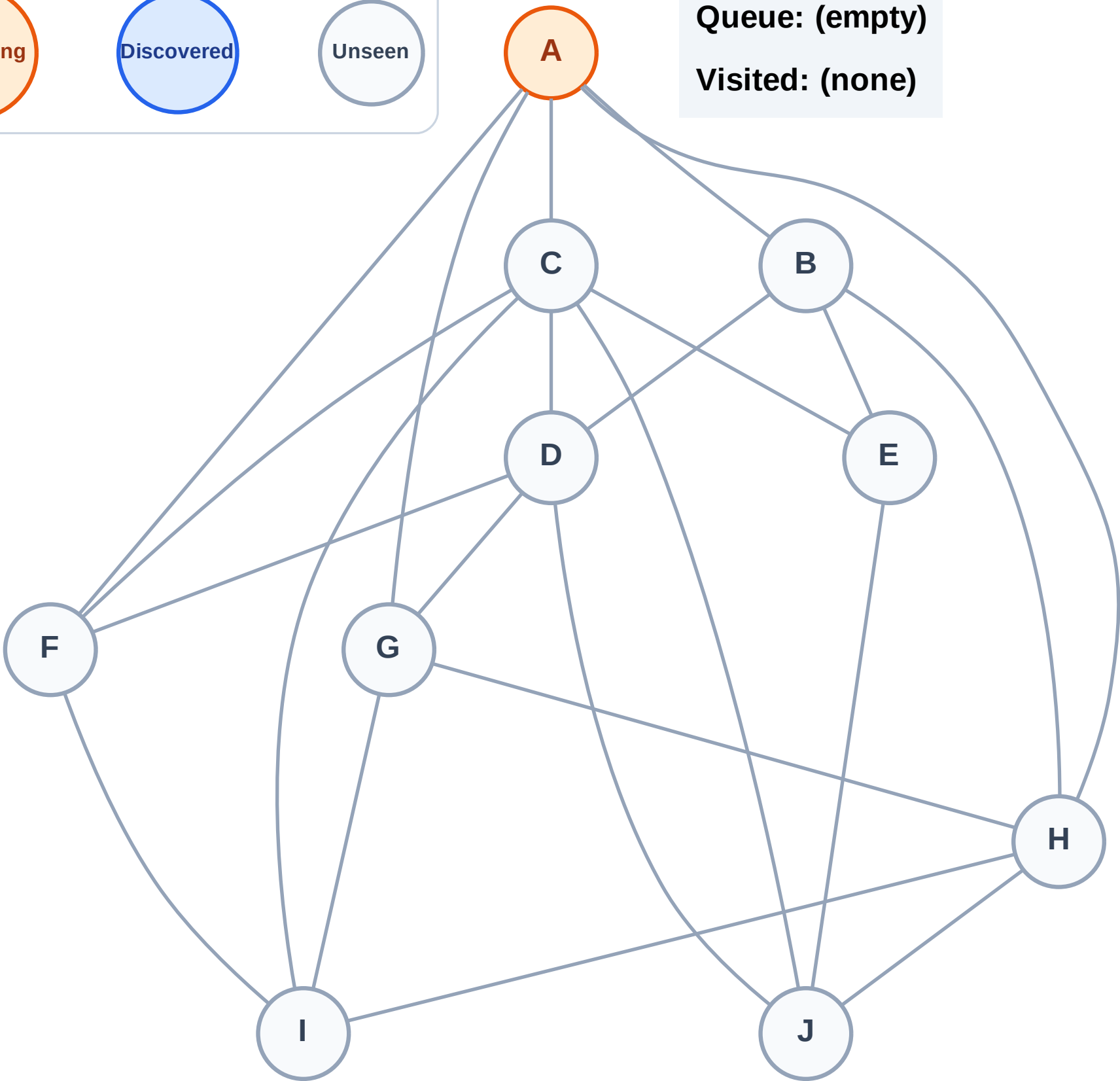
Complete

Processing

Discovered

Unseen

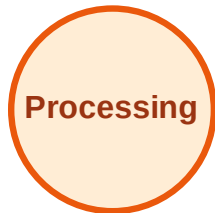
Queue: (empty)  
Visited: (none)



# BFS Traversal

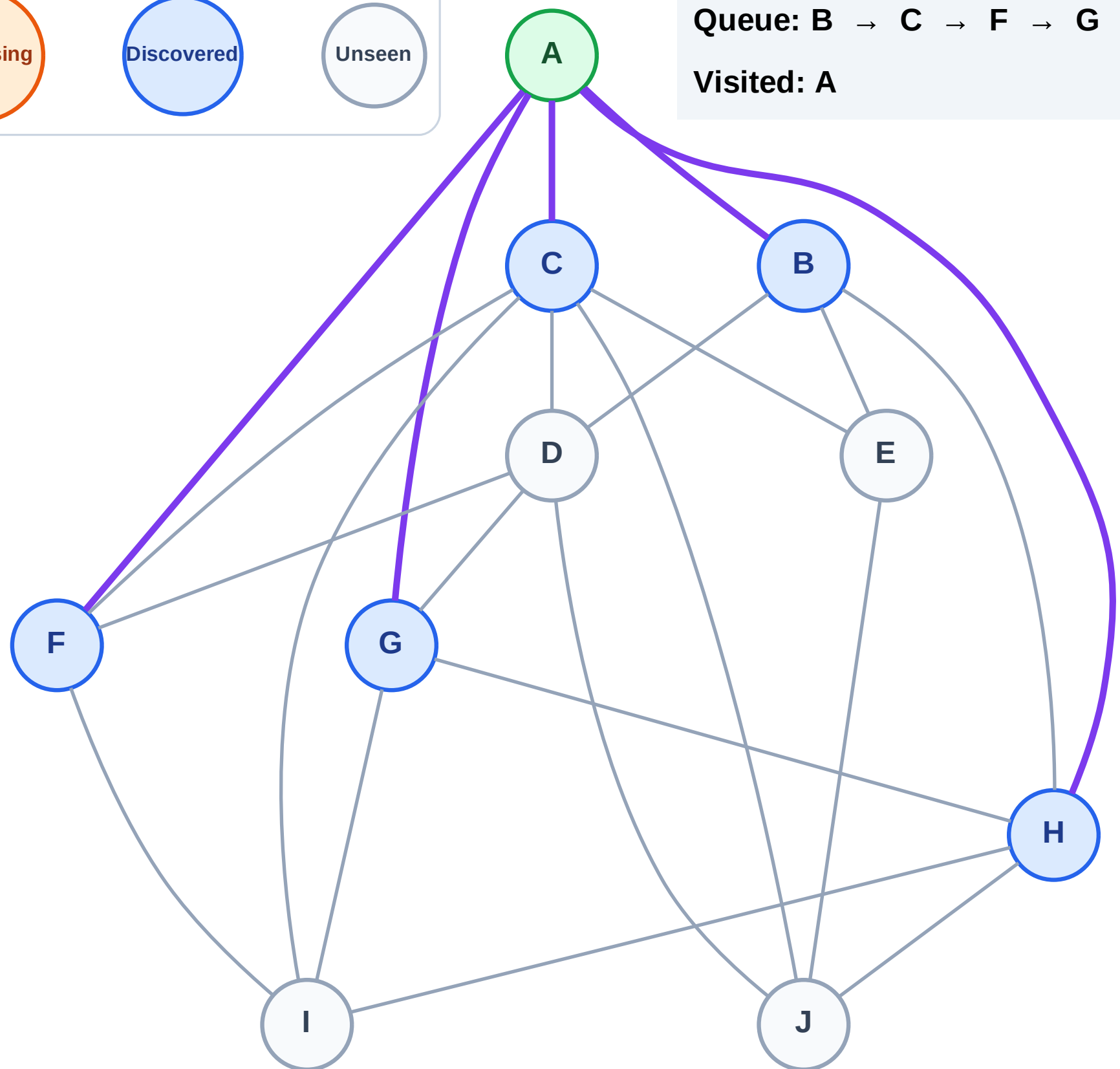
Step 3: Discover B, C, F, G, H from A

## Legend



Queue: B → C → F → G → H

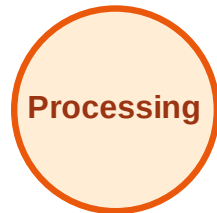
Visited: A



# BFS Traversal

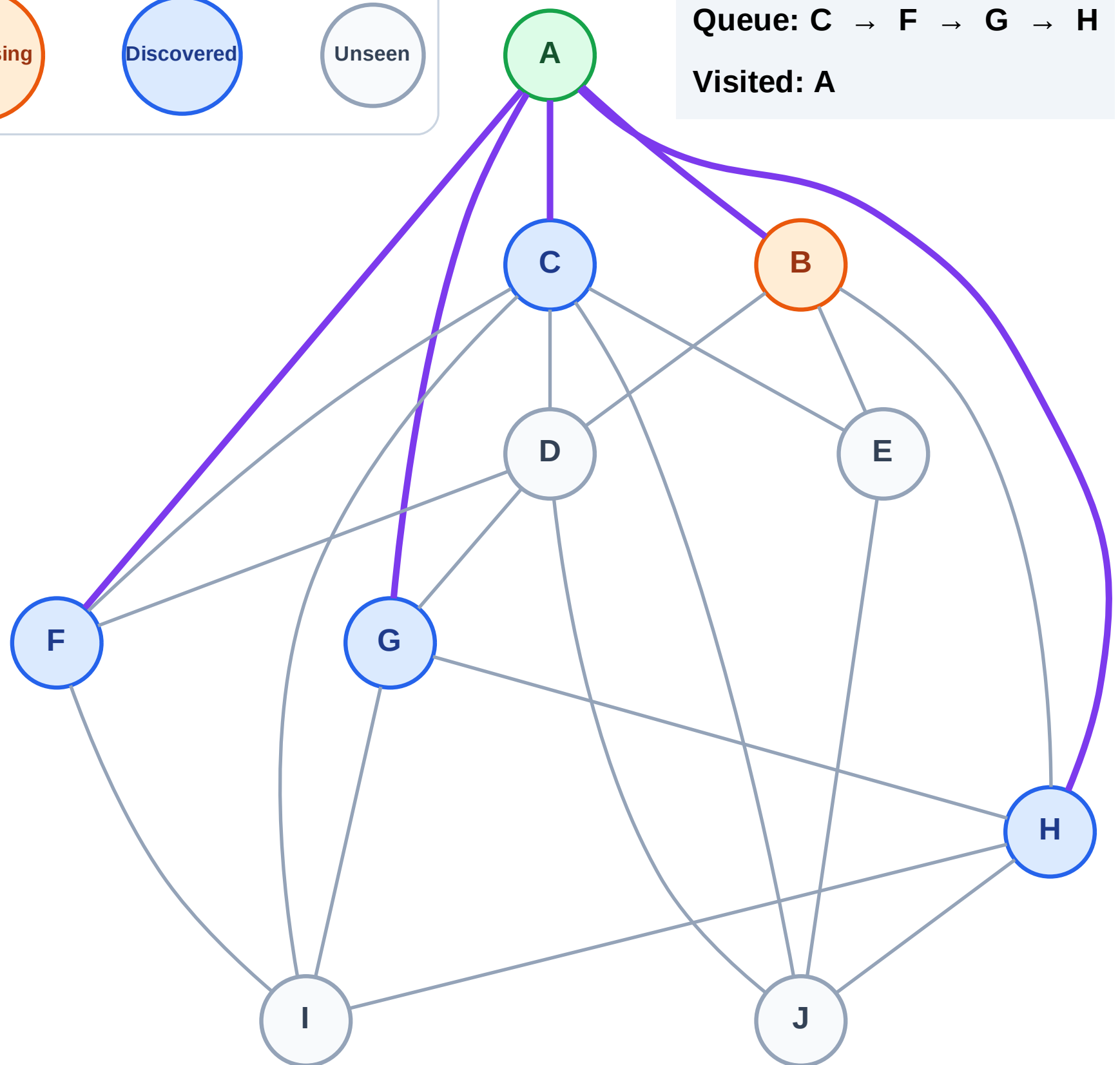
Step 4: Process node B

## Legend



Queue: C → F → G → H

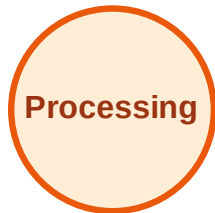
Visited: A



# BFS Traversal

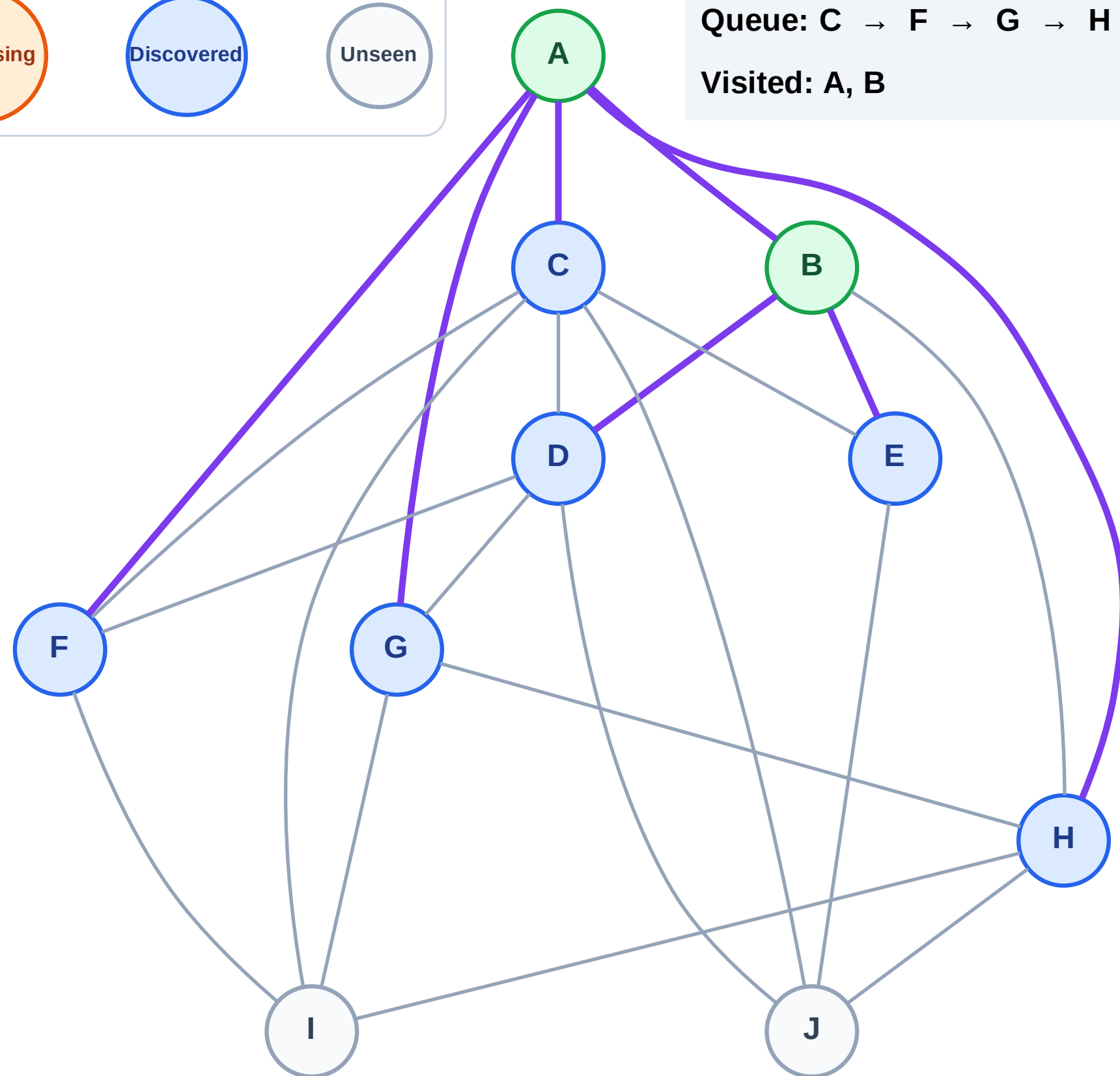
Step 5: Discover D, E from B

## Legend



Queue: C → F → G → H → D → E

Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

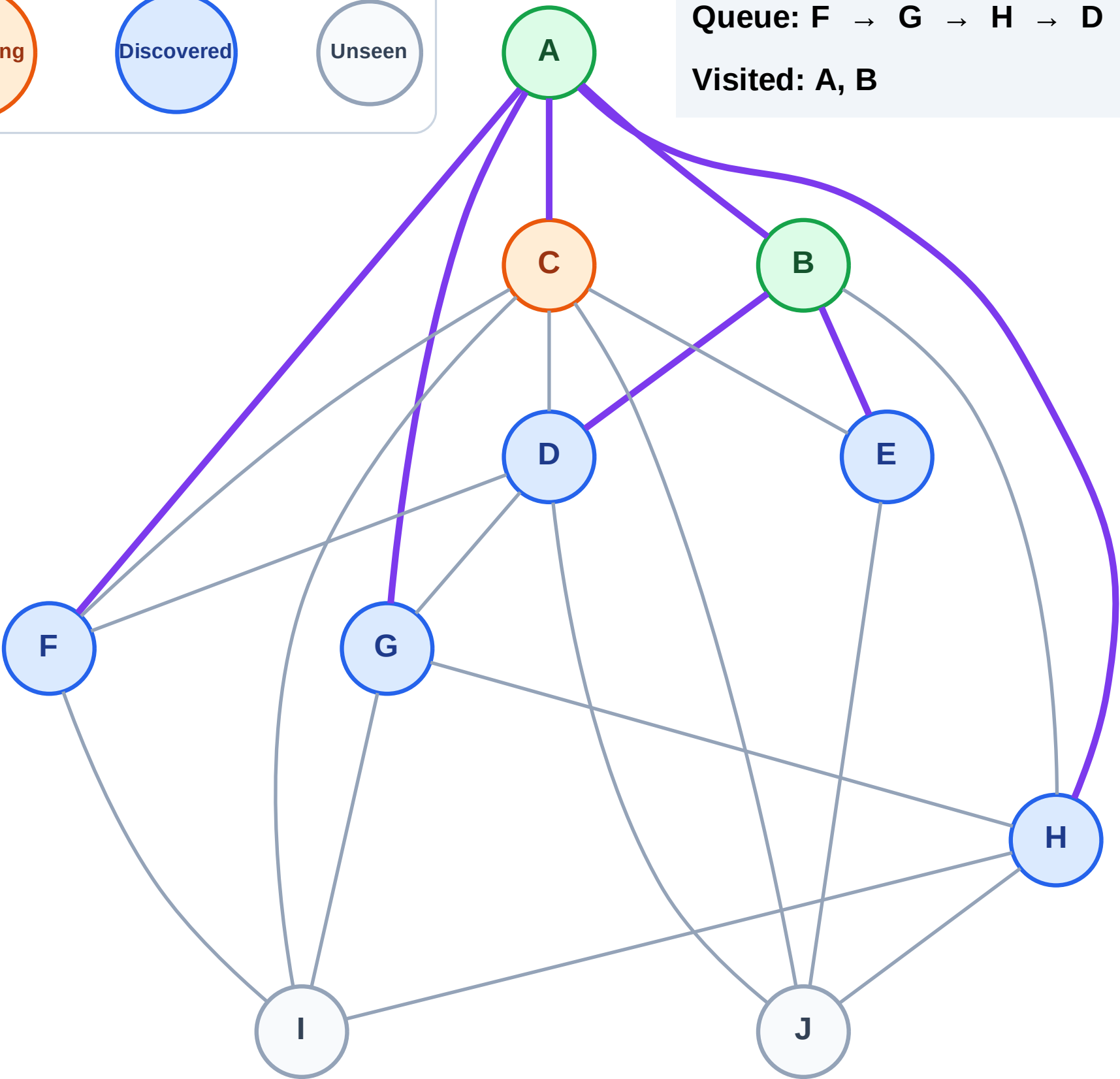
Processing

Discovered

Unseen

Queue: F → G → H → D → E

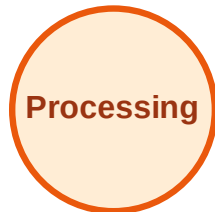
Visited: A, B



# BFS Traversal

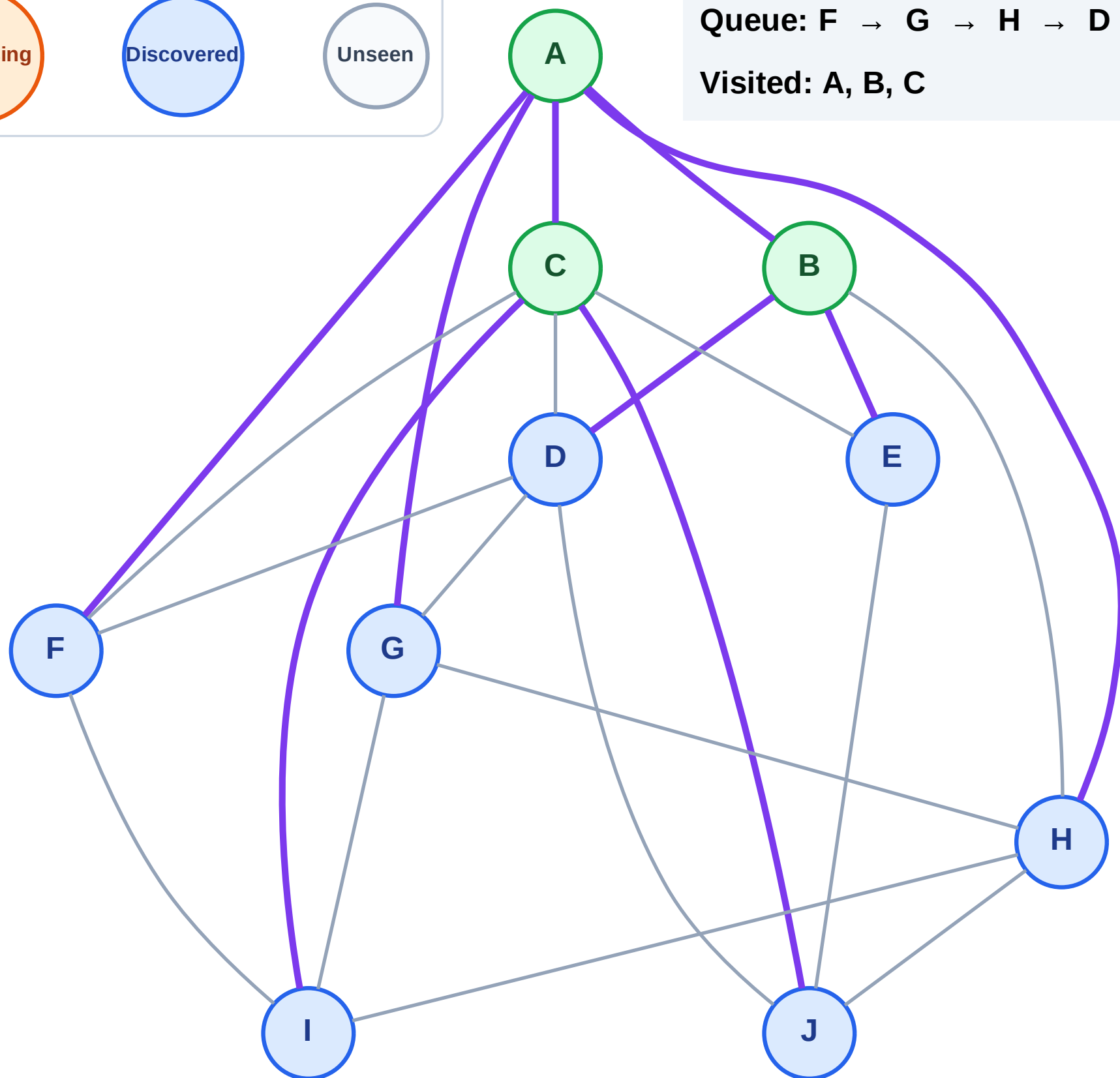
Step 7: Discover I, J from C

## Legend



Queue: F → G → H → D → E → I → J

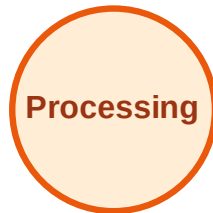
Visited: A, B, C



# BFS Traversal

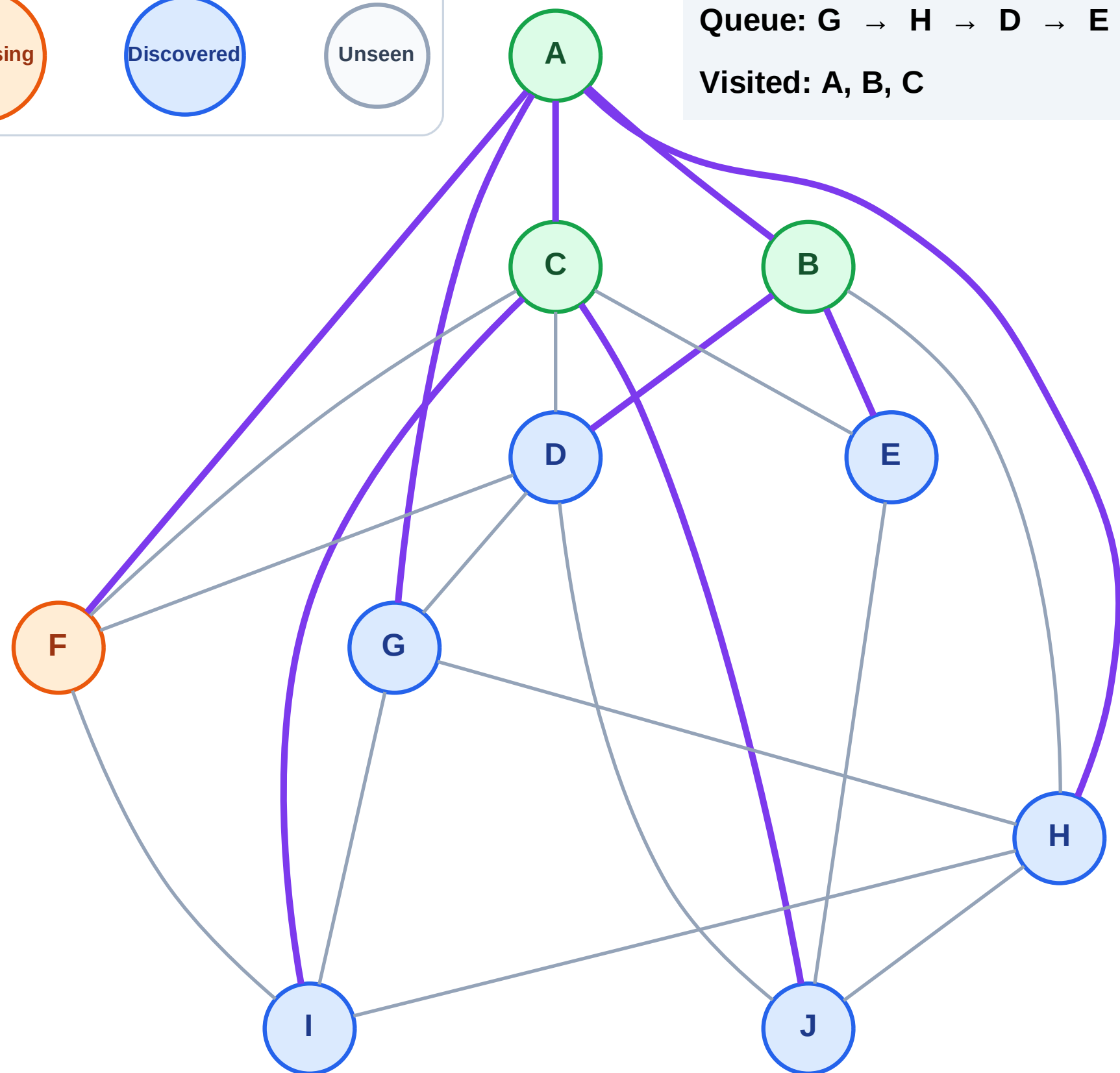
Step 8: Process node F

## Legend



Queue: G → H → D → E → I → J

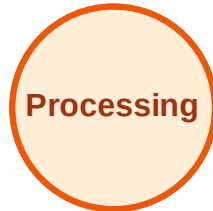
Visited: A, B, C



# BFS Traversal

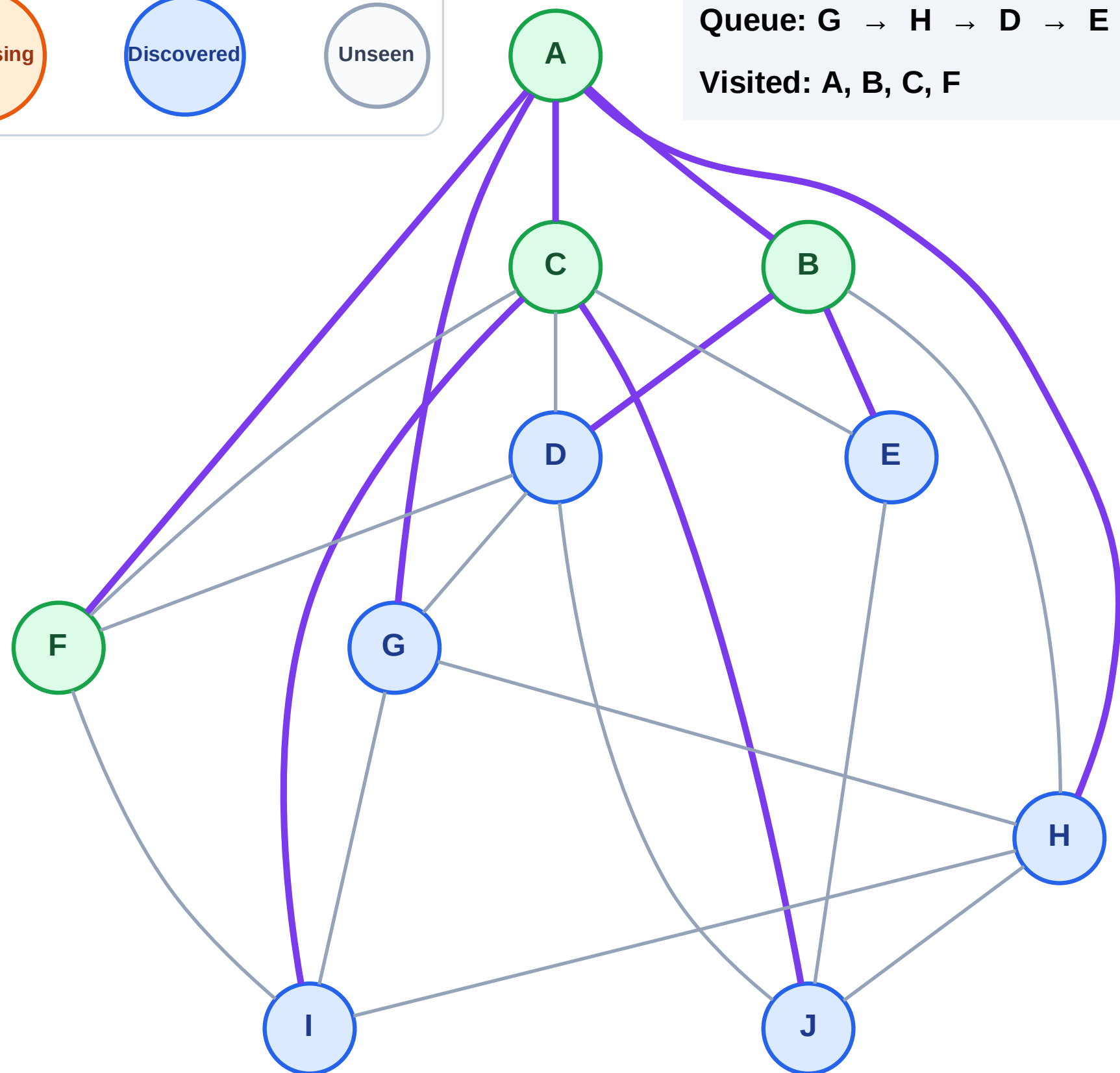
Step 9: No new neighbors from F

## Legend



Queue: G → H → D → E → I → J

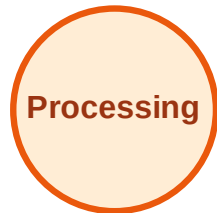
Visited: A, B, C, F



# BFS Traversal

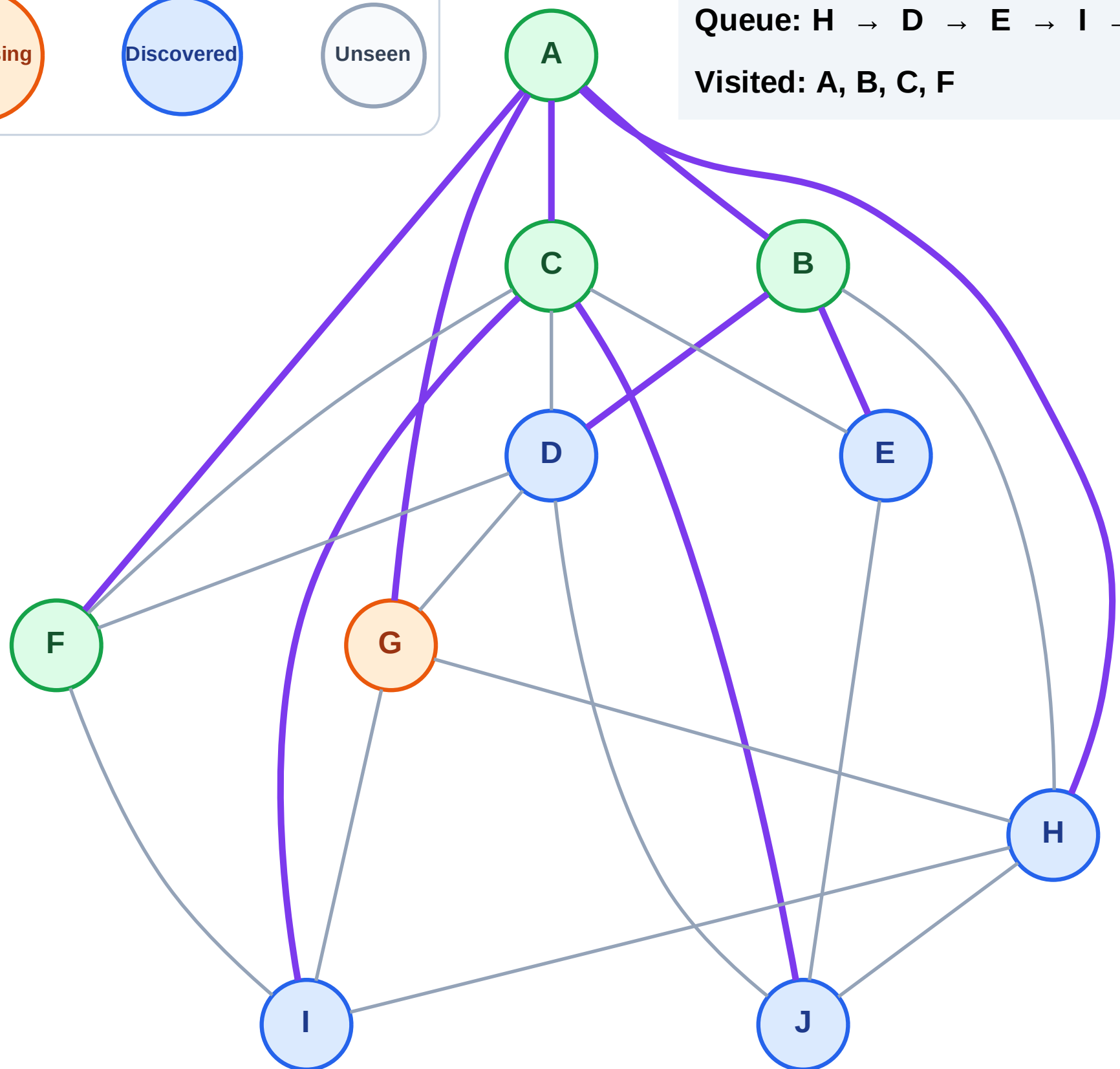
Step 10: Process node G

## Legend



Queue: H → D → E → I → J

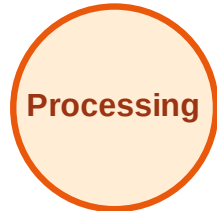
Visited: A, B, C, F



# BFS Traversal

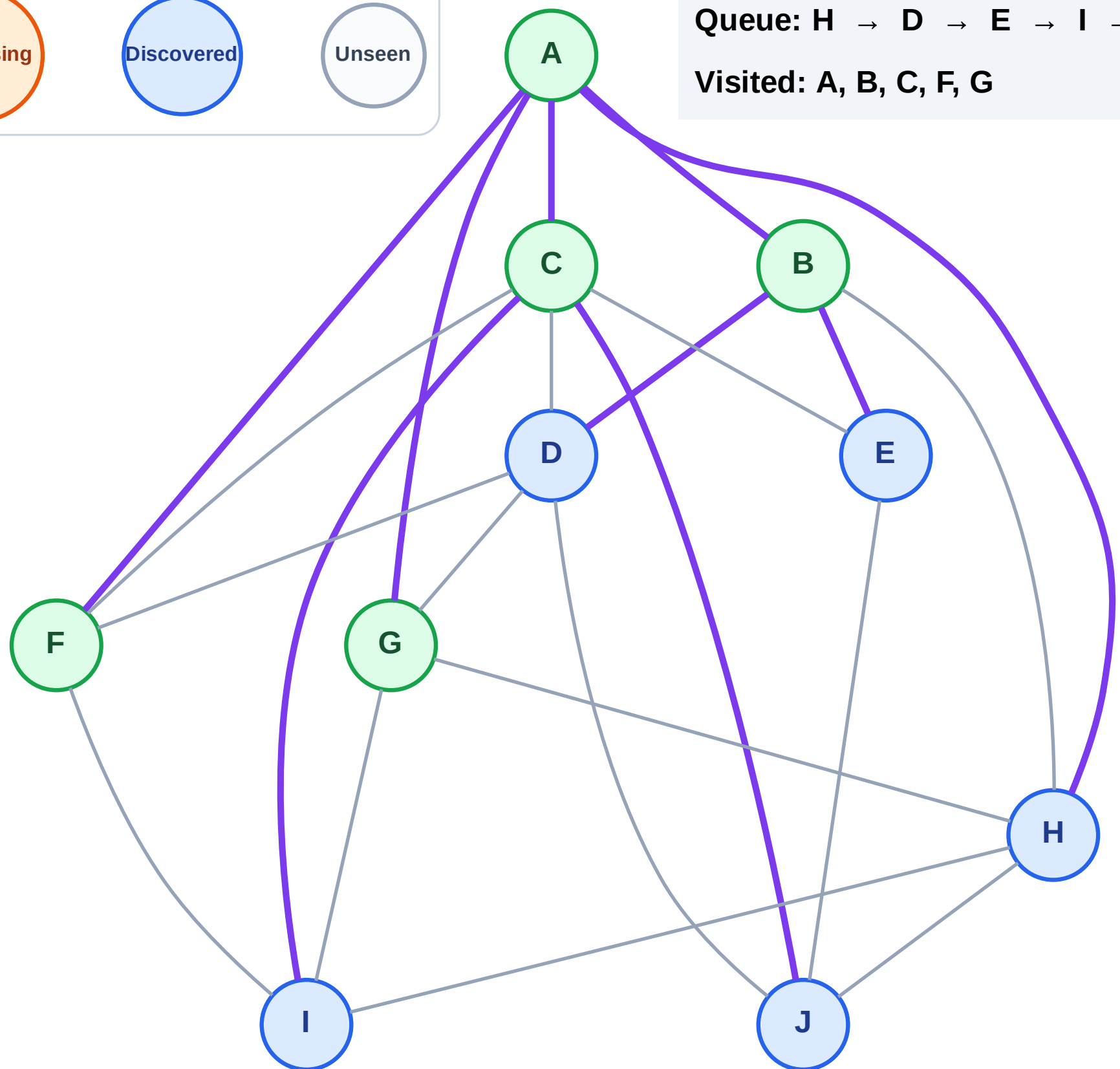
Step 11: No new neighbors from G

## Legend



Queue: H → D → E → I → J

Visited: A, B, C, F, G



# BFS Traversal

Step 12: Process node H

Legend

Complete

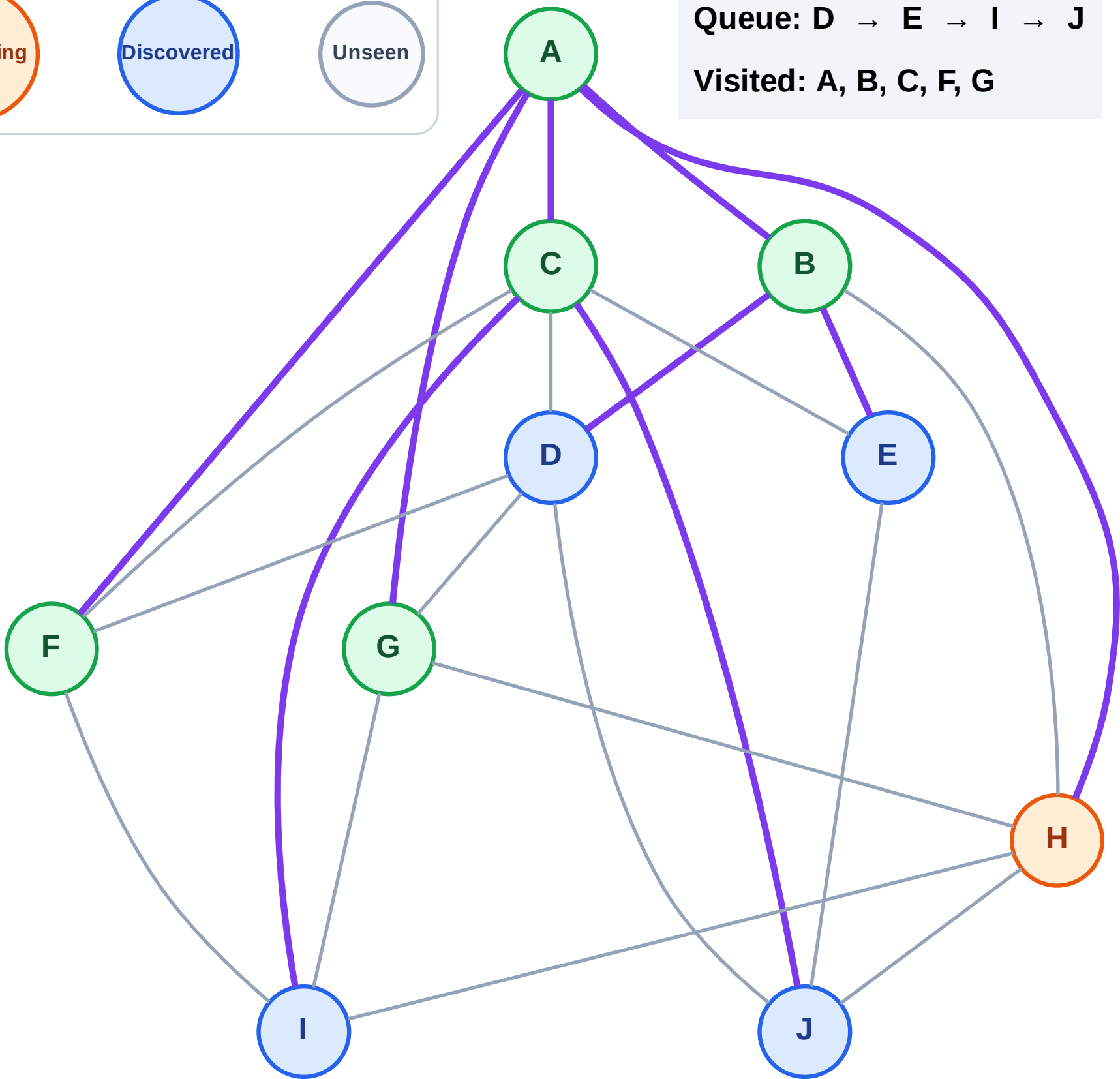
Processing

Discovered

Unseen

Queue: D → E → I → J

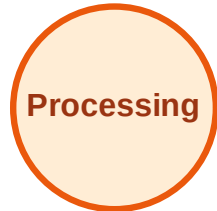
Visited: A, B, C, F, G



# BFS Traversal

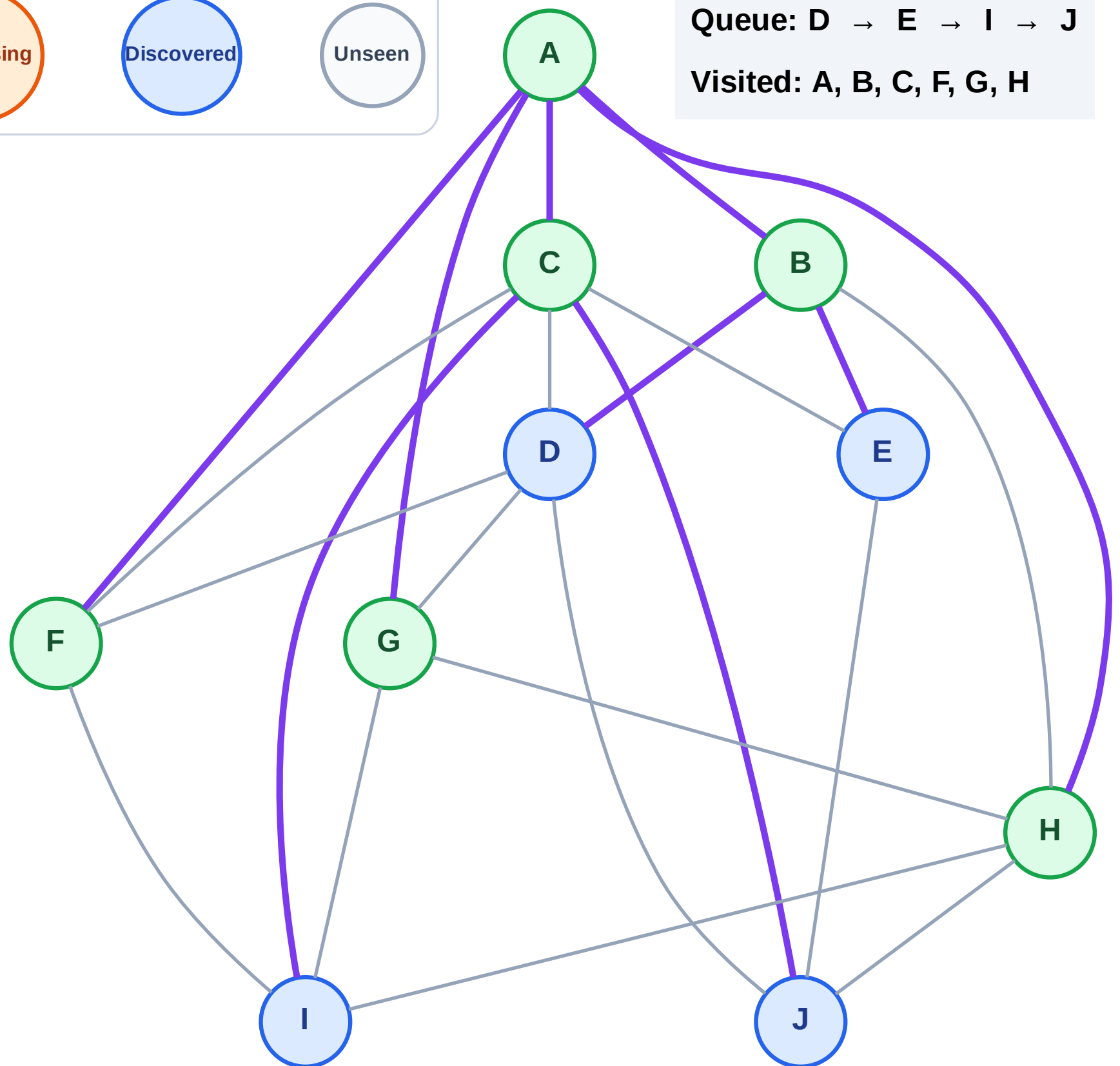
Step 13: No new neighbors from H

## Legend



Queue: D → E → I → J

Visited: A, B, C, F, G, H



# BFS Traversal

Step 14: Process node D

Legend

Complete

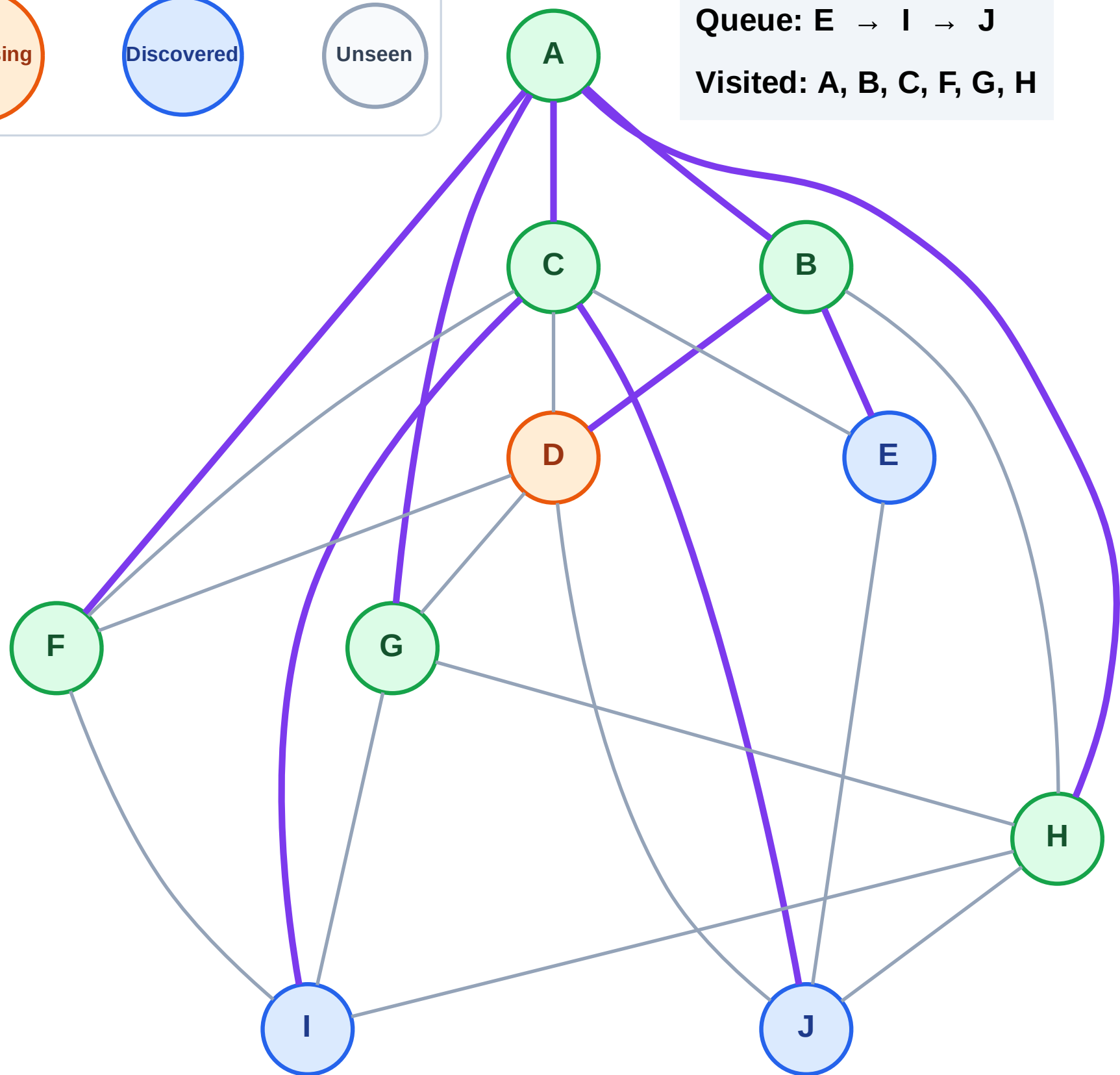
Processing

Discovered

Unseen

Queue: E → I → J

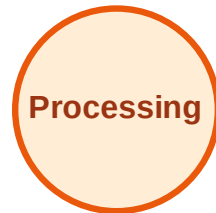
Visited: A, B, C, F, G, H



# BFS Traversal

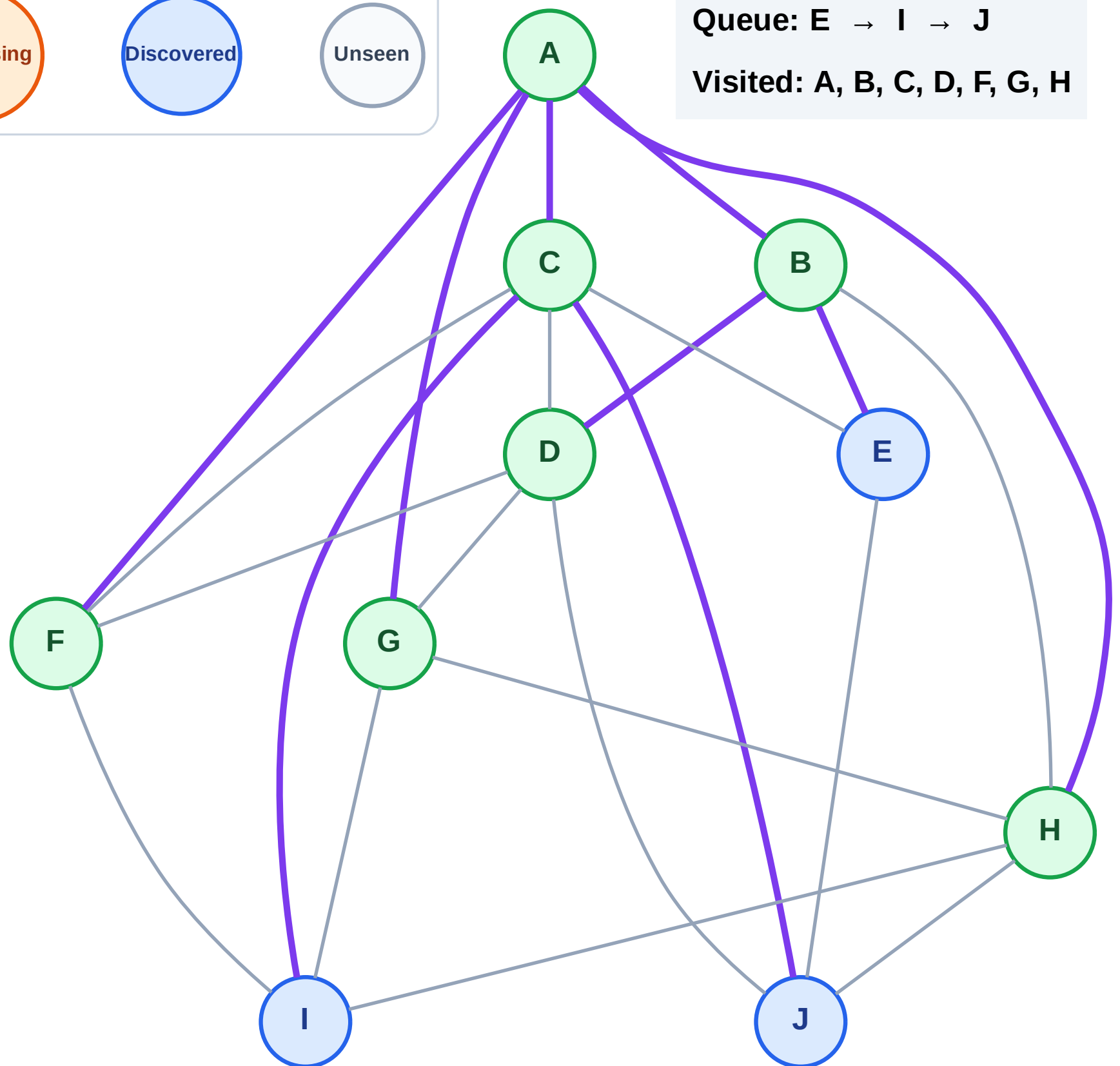
Step 15: No new neighbors from D

## Legend



Queue: E → I → J

Visited: A, B, C, D, F, G, H



# BFS Traversal

Step 16: Process node E

Legend

Complete

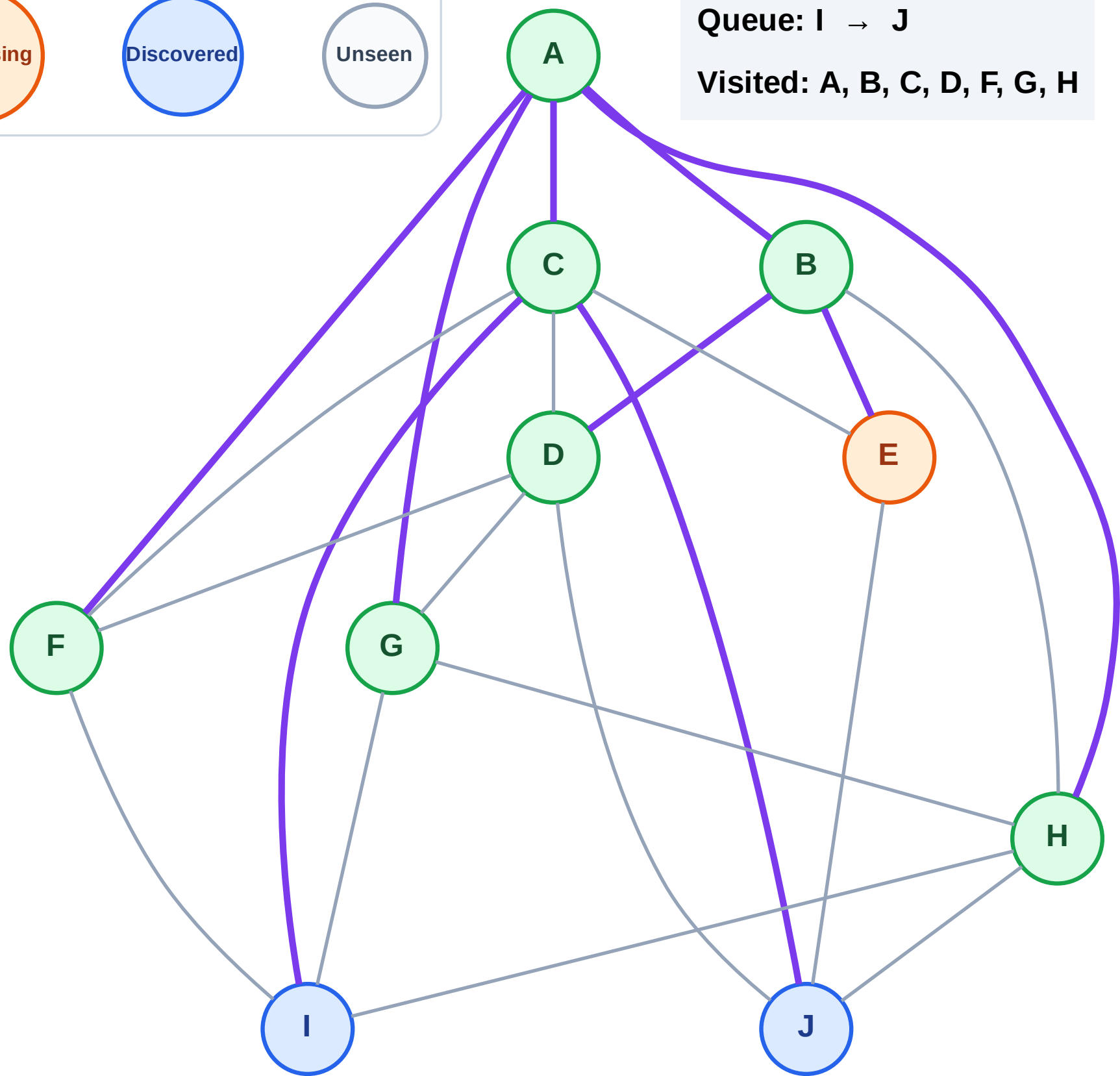
Processing

Discovered

Unseen

Queue: I → J

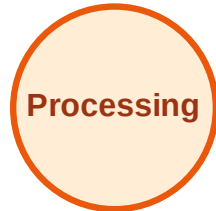
Visited: A, B, C, D, F, G, H



# BFS Traversal

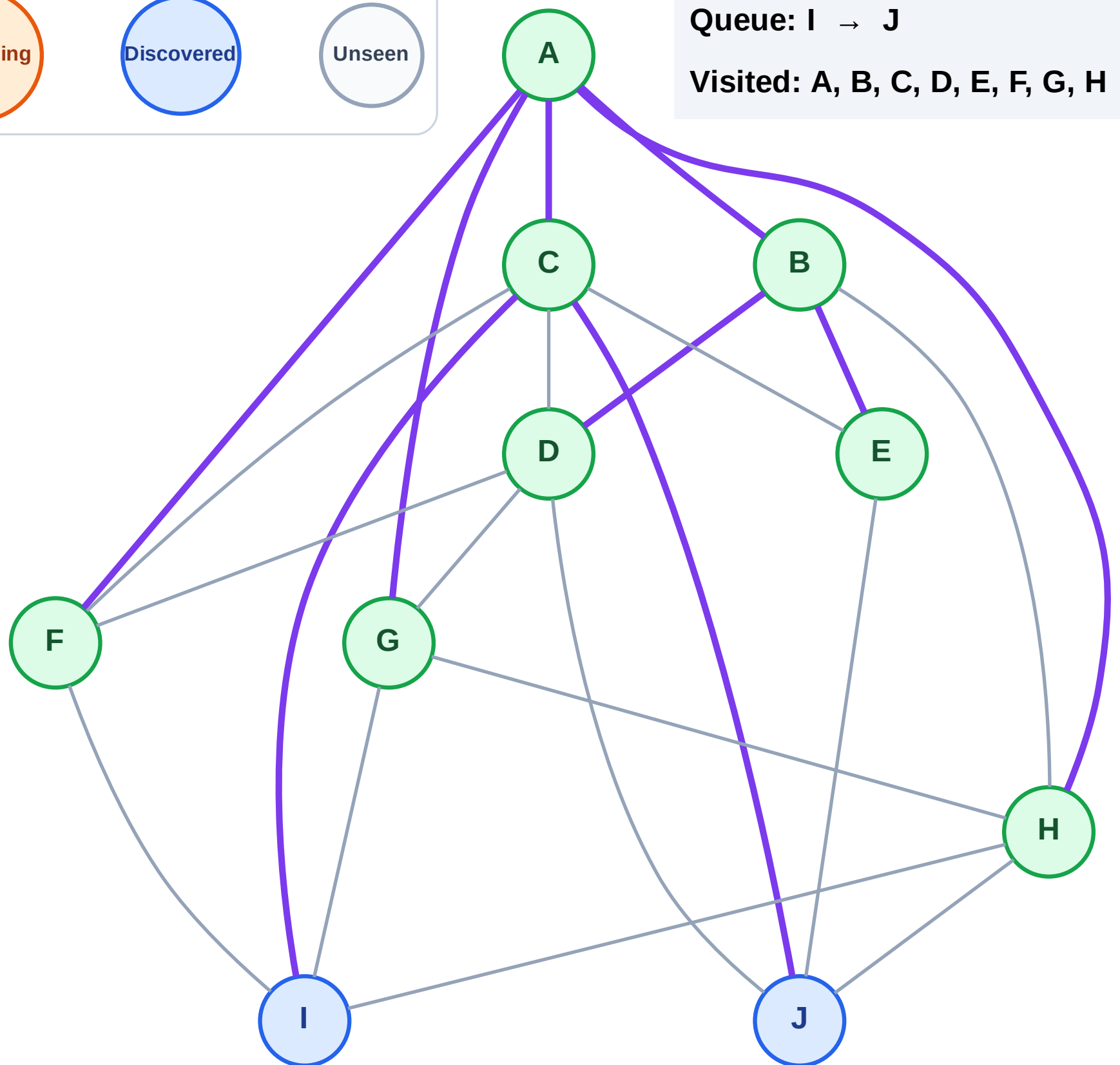
Step 17: No new neighbors from E

## Legend



Queue: I → J

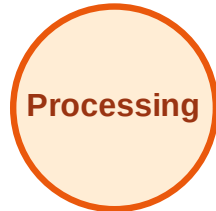
Visited: A, B, C, D, E, F, G, H



# BFS Traversal

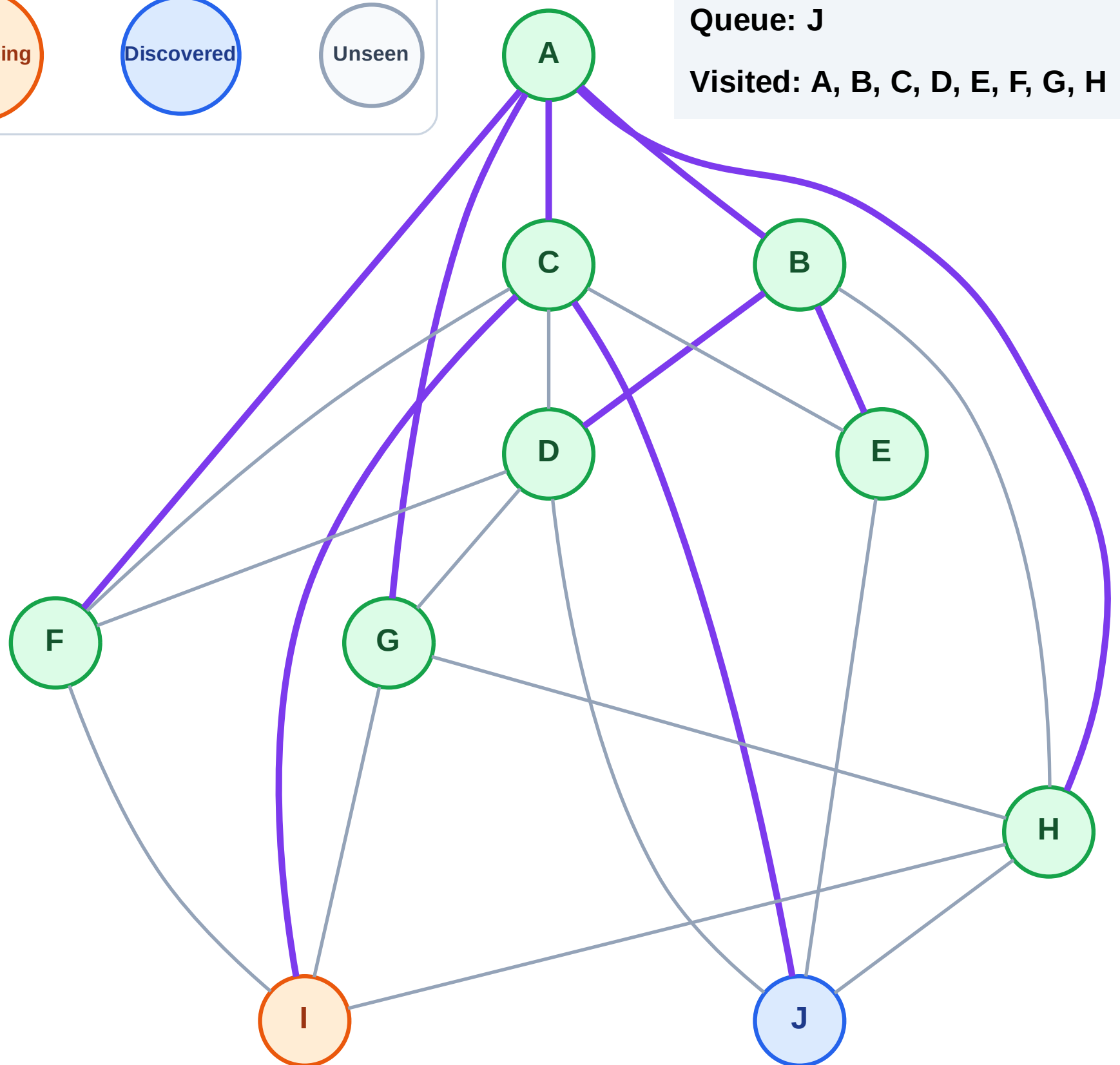
Step 18: Process node I

## Legend



Queue: J

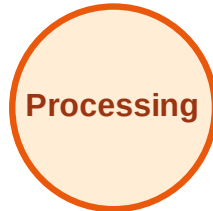
Visited: A, B, C, D, E, F, G, H



# BFS Traversal

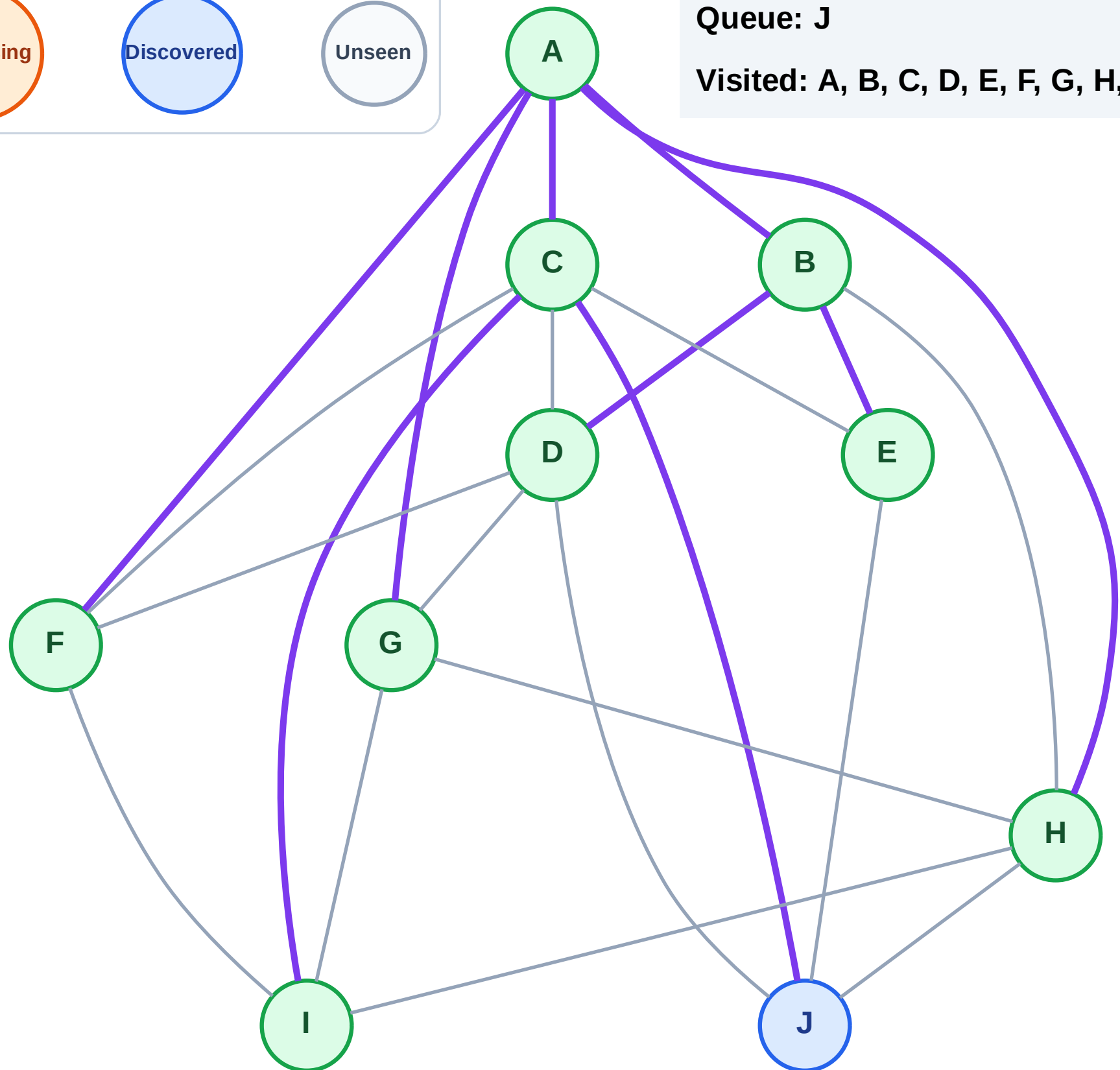
Step 19: No new neighbors from I

## Legend



Queue: J

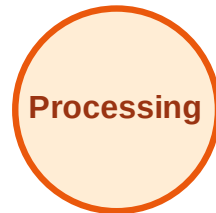
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

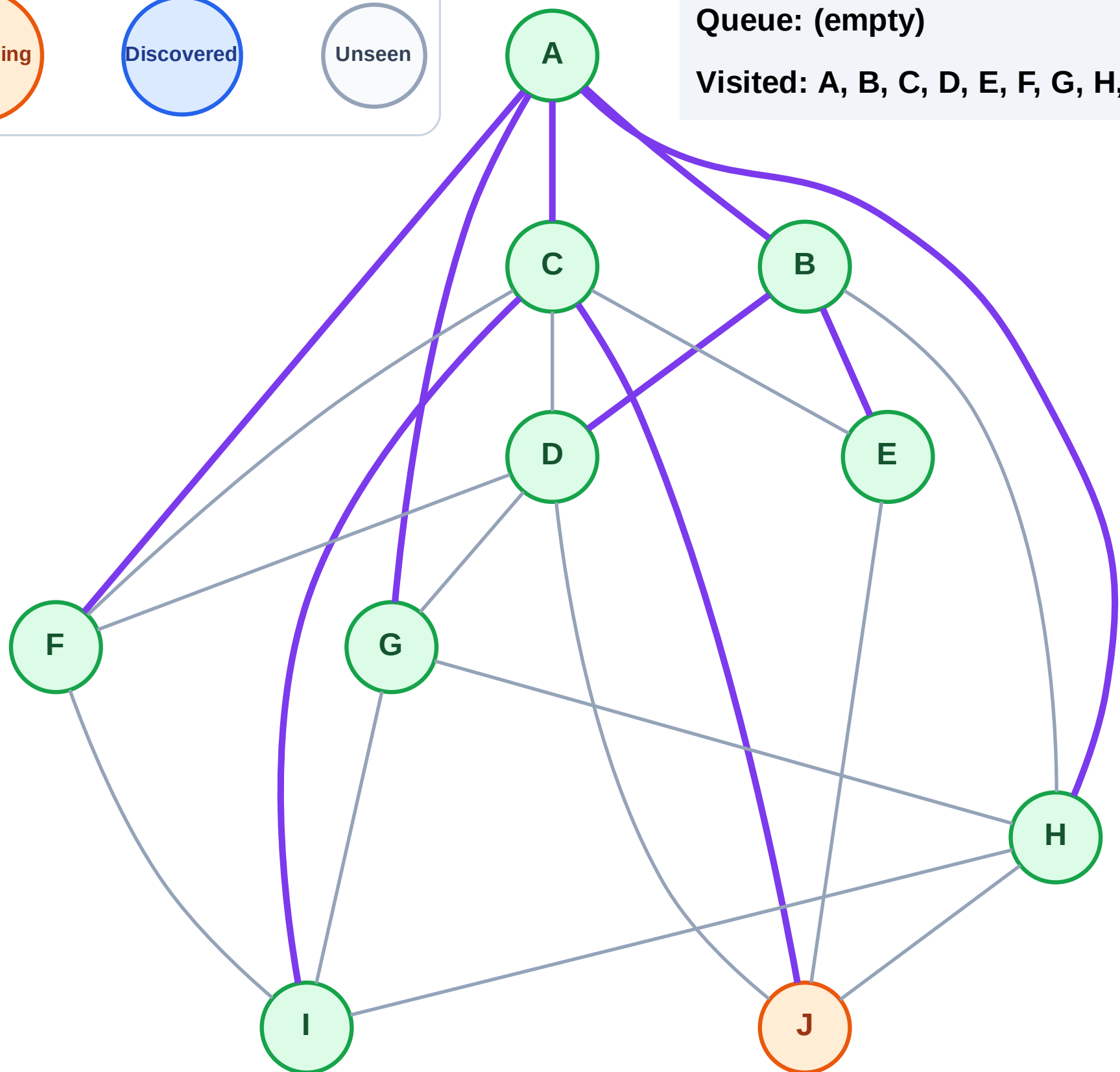
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

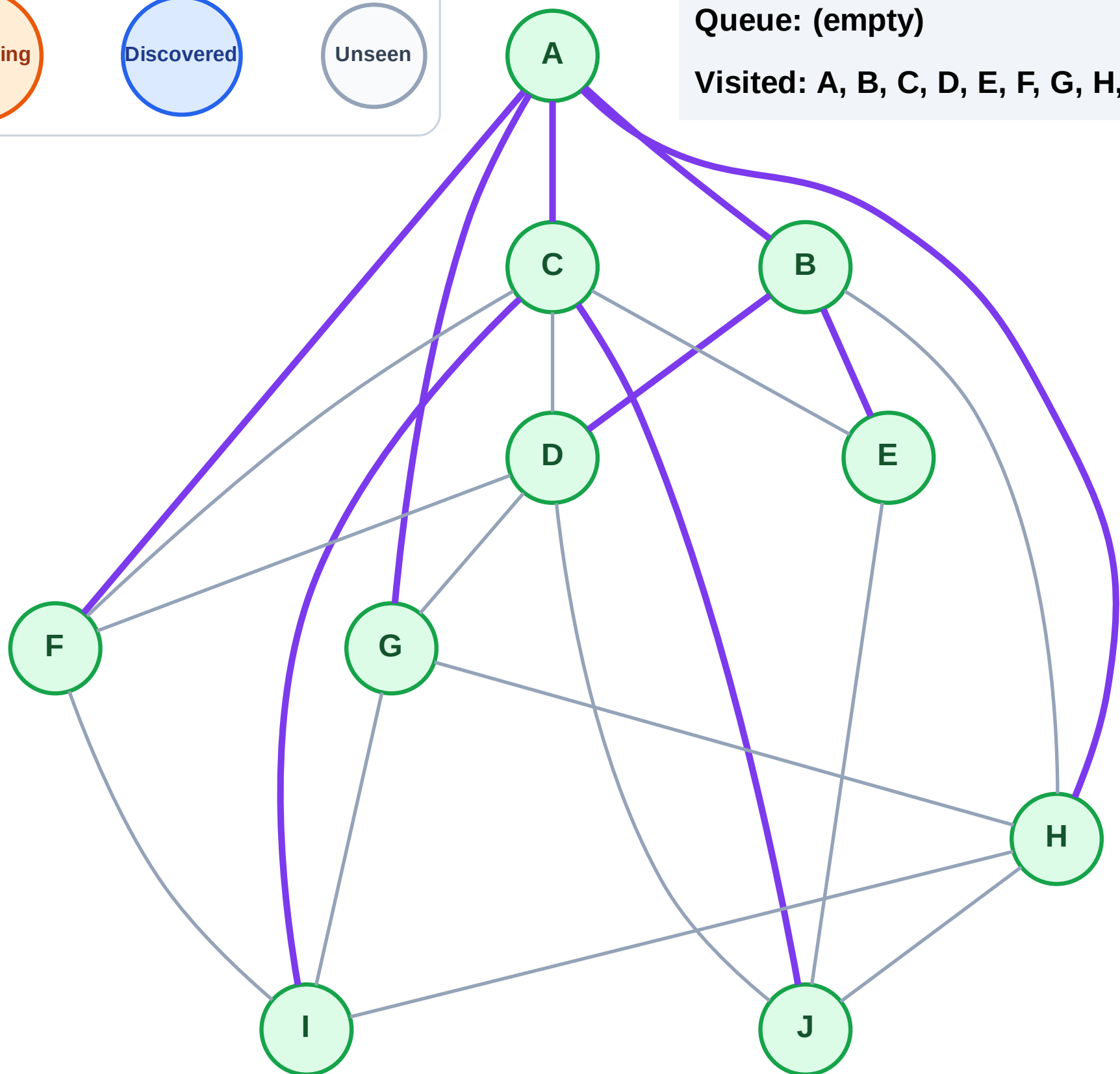
Step 21: No new neighbors from J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

Step 1: Start at A

Legend

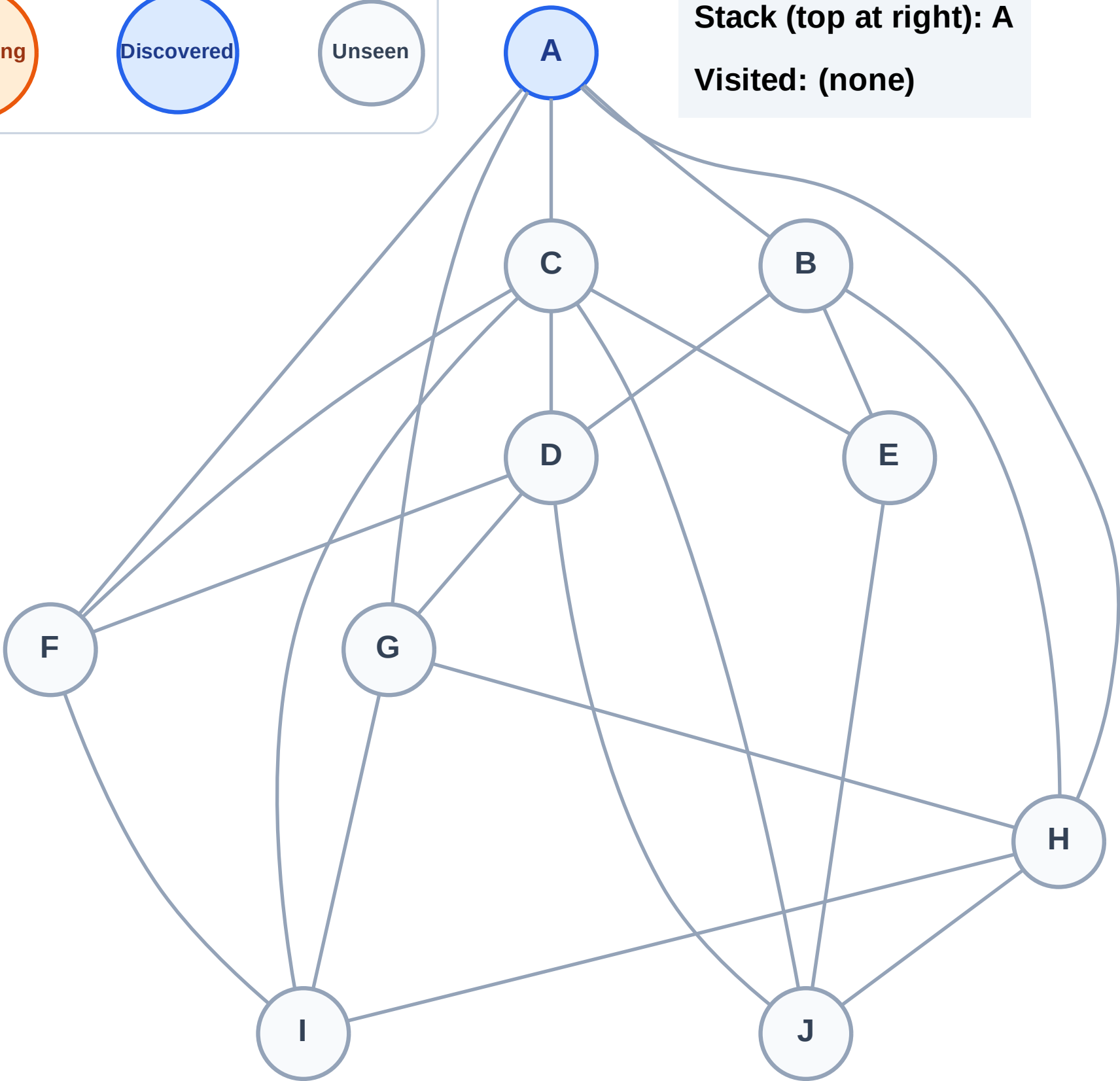
Complete

Processing

Discovered

Unseen

Stack (top at right): A  
Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

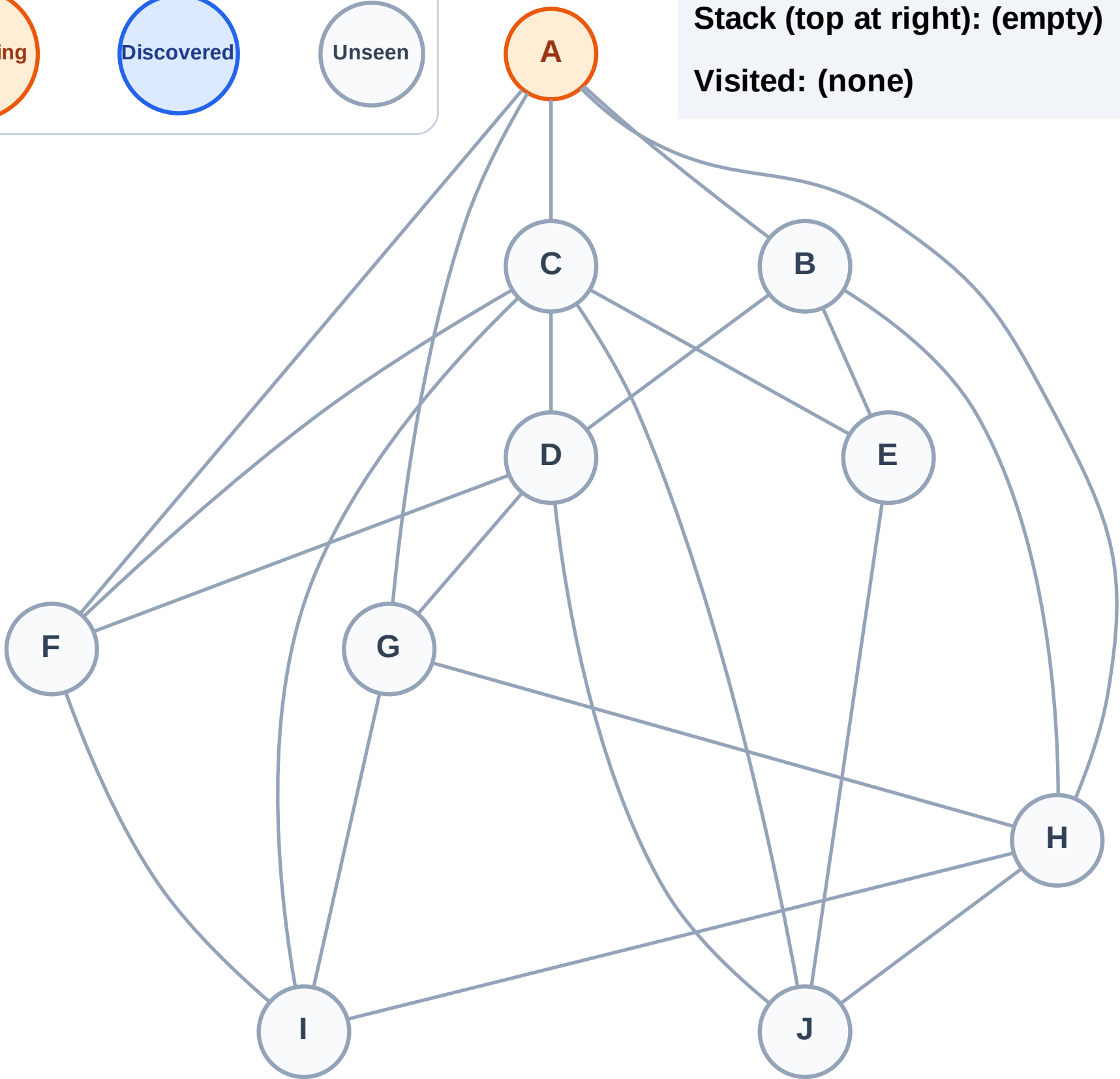
Complete

Processing

Discovered

Unseen

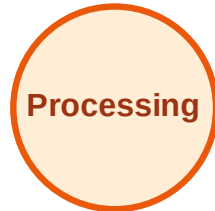
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

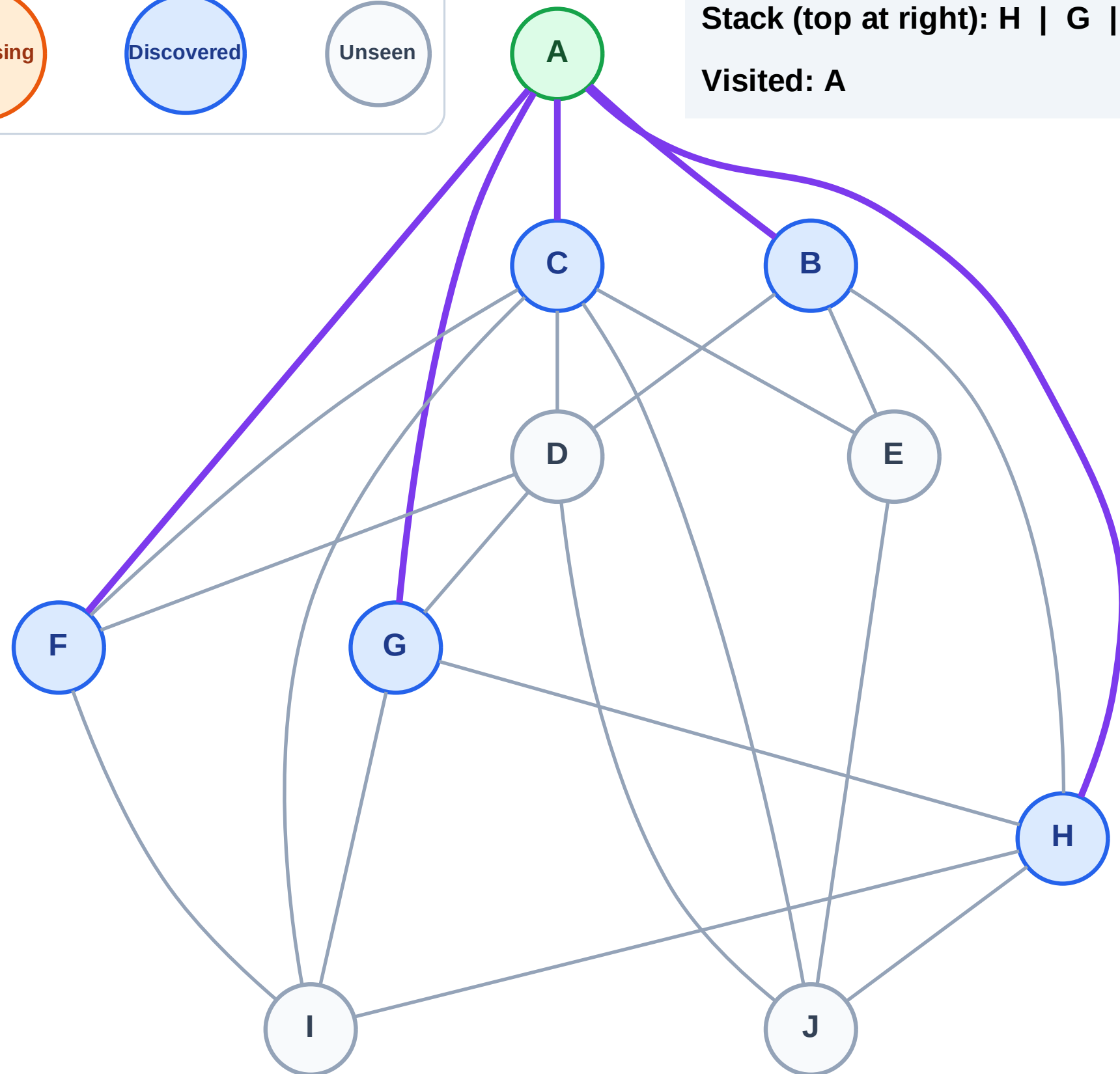
Step 3: Discover H, G, F, C, B from A

## Legend



Stack (top at right): H | G | F | C | B

Visited: A



# DFS Traversal

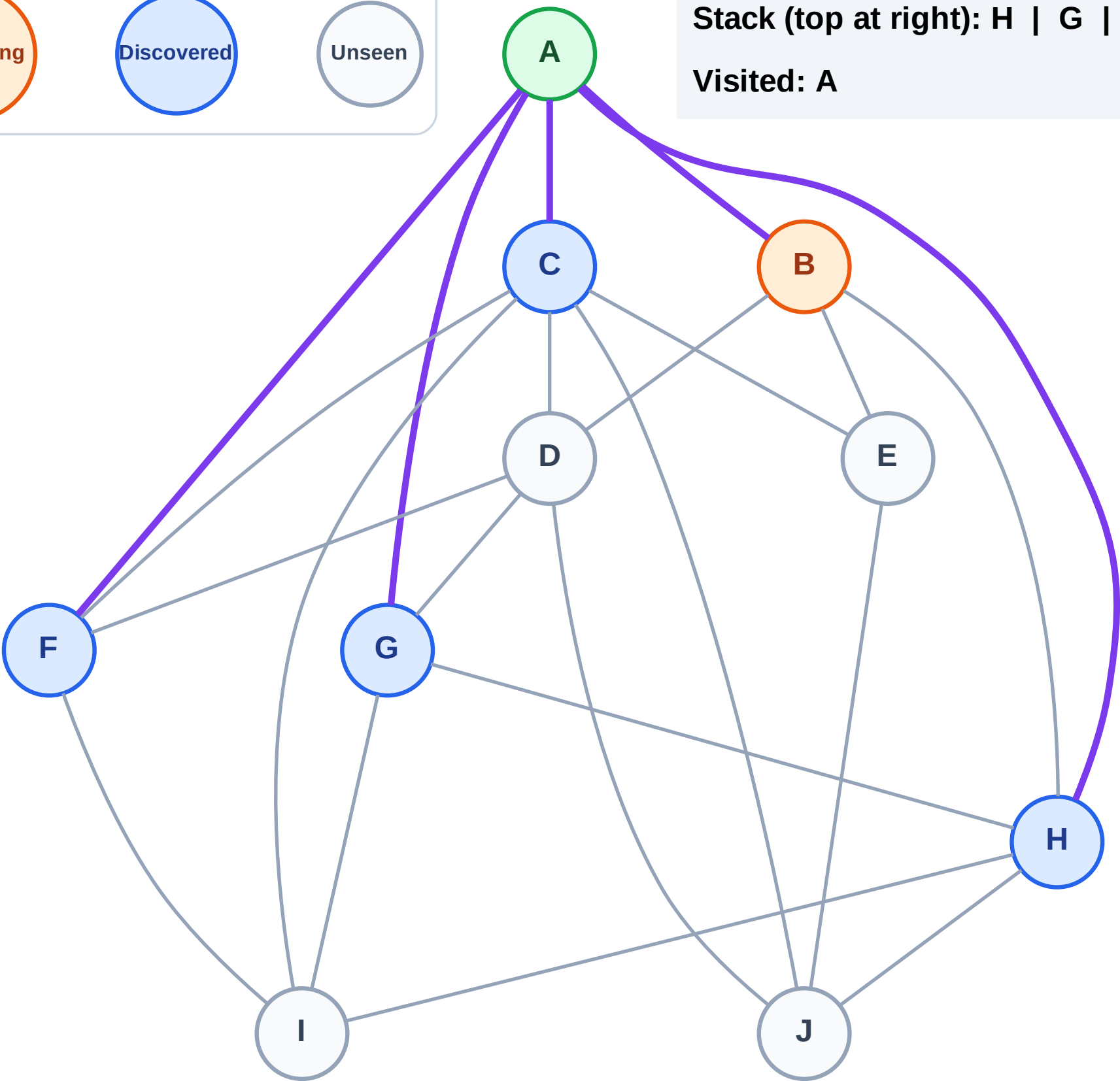
Step 4: Process node B

## Legend



Stack (top at right): H | G | F | C

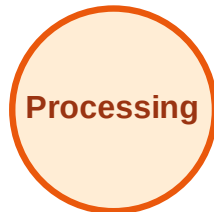
Visited: A



# DFS Traversal

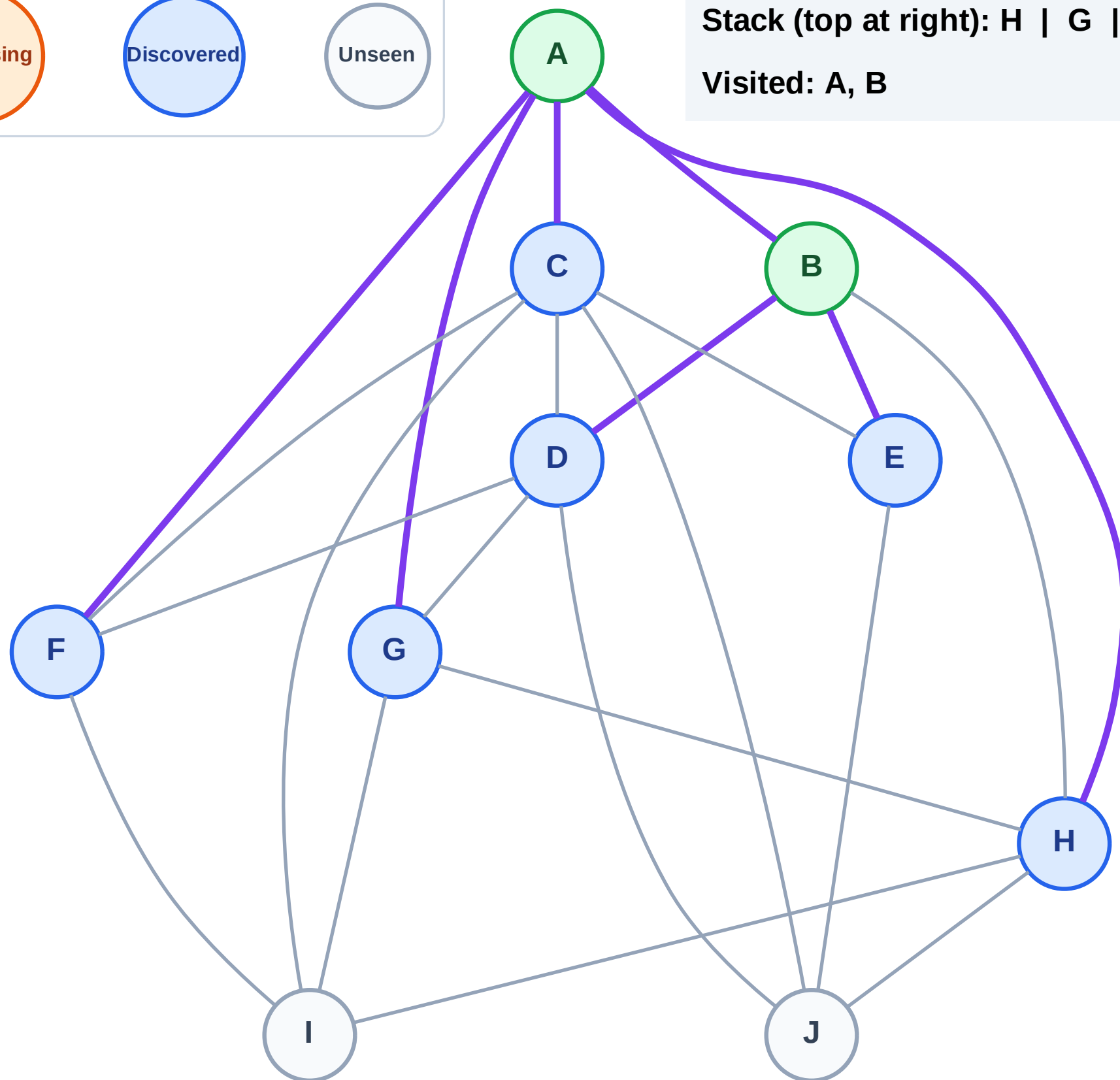
Step 5: Discover E, D from B

## Legend



Stack (top at right): H | G | F | C | E | D

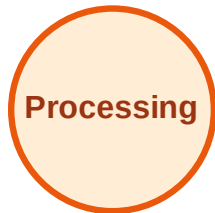
Visited: A, B



# DFS Traversal

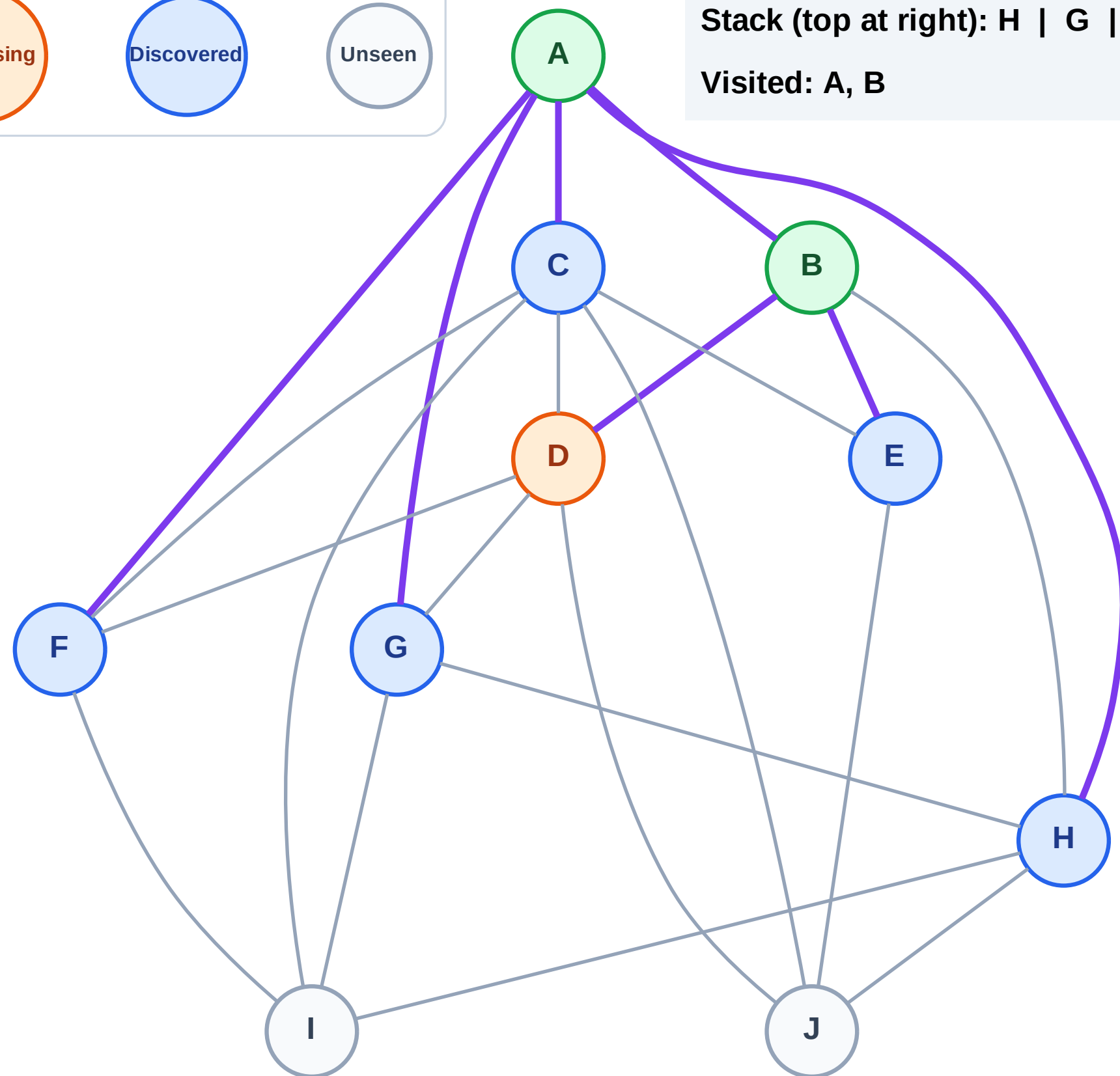
Step 6: Process node D

## Legend



Stack (top at right): H | G | F | C | E

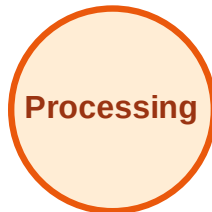
Visited: A, B



# DFS Traversal

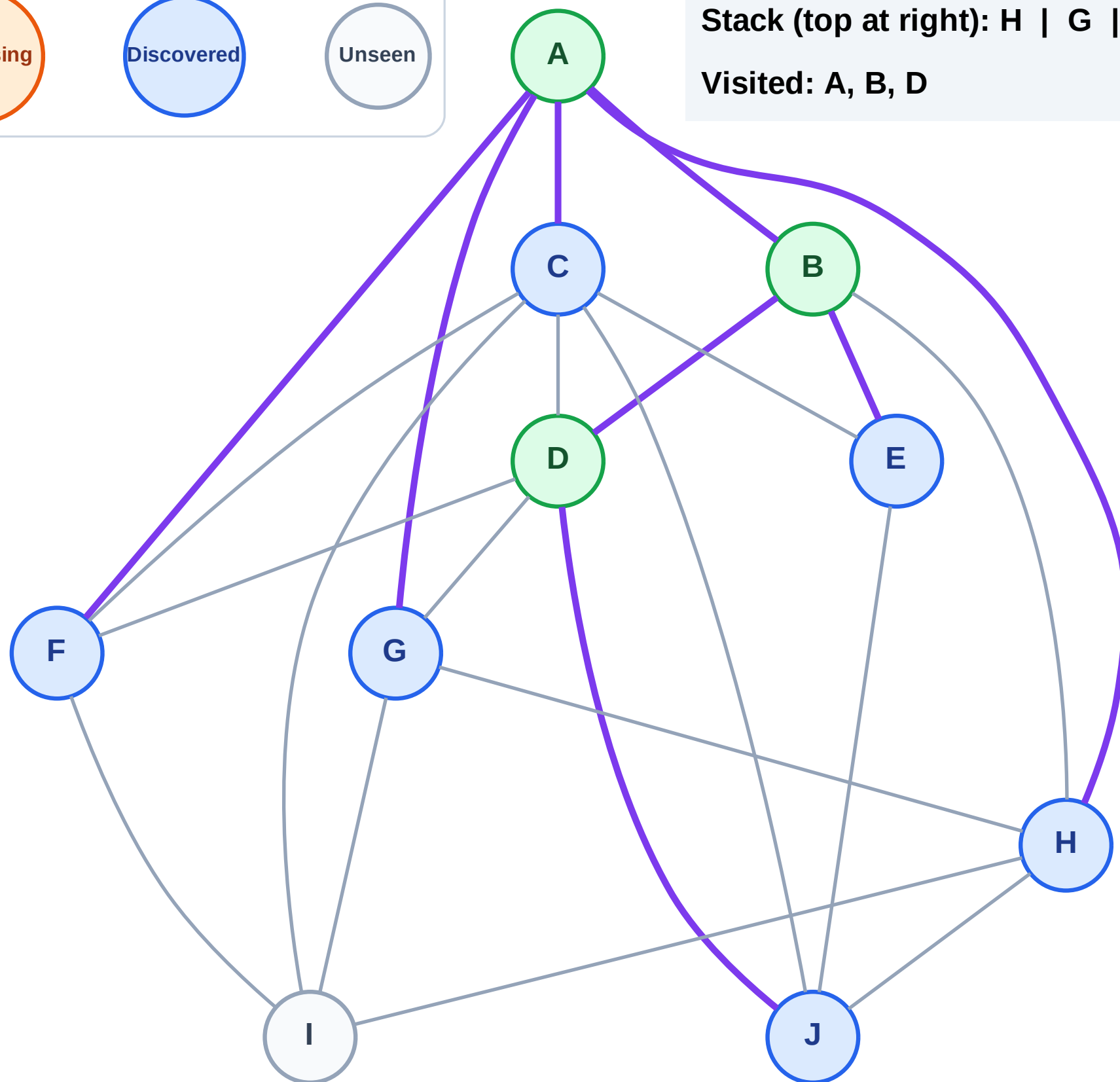
Step 7: Discover J from D

## Legend



Stack (top at right): H | G | F | C | E | J

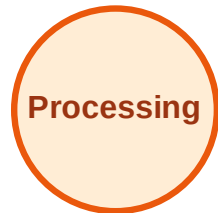
Visited: A, B, D



# DFS Traversal

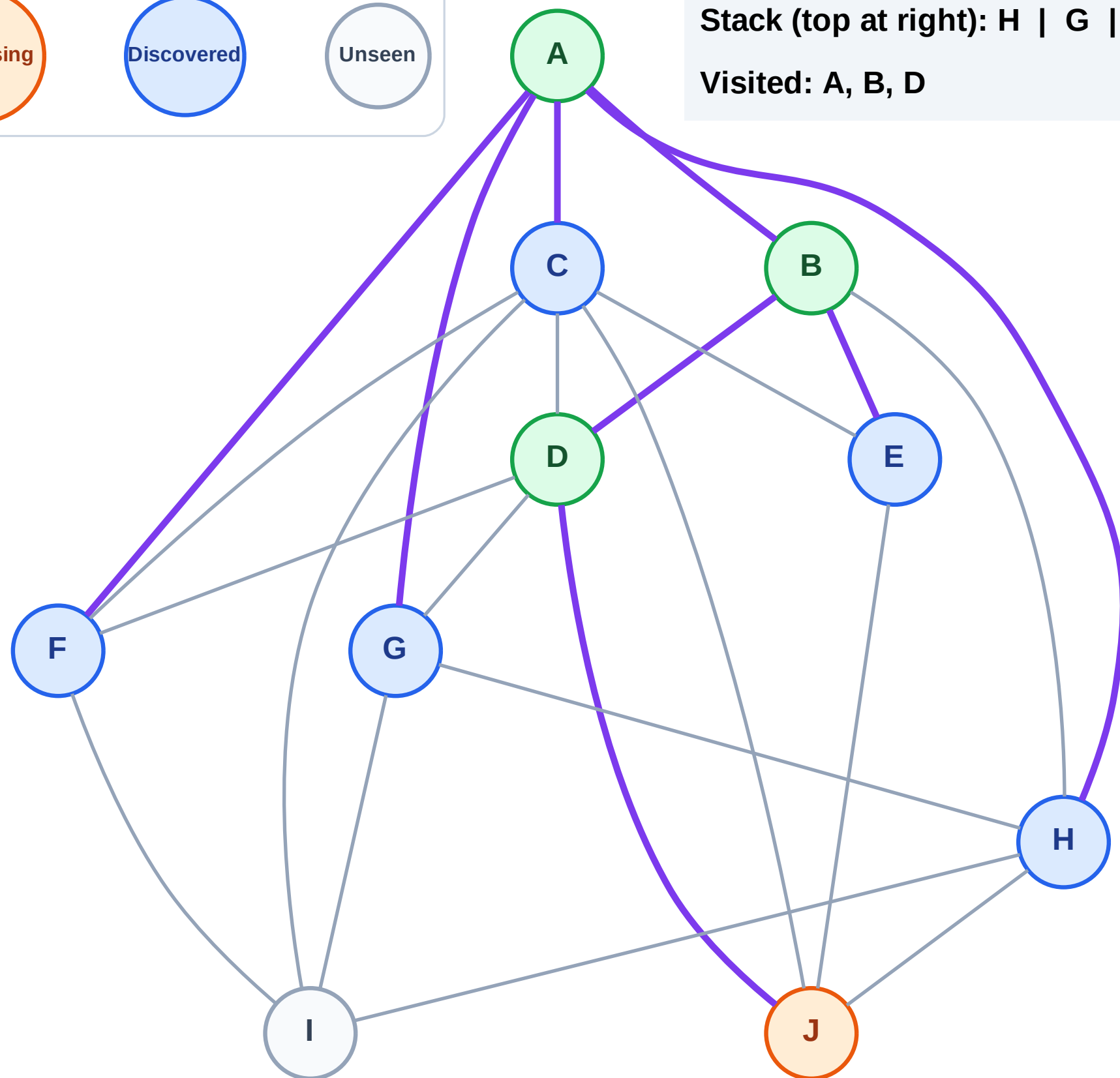
Step 8: Process node J

## Legend



Stack (top at right): H | G | F | C | E

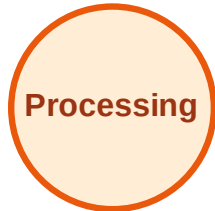
Visited: A, B, D



# DFS Traversal

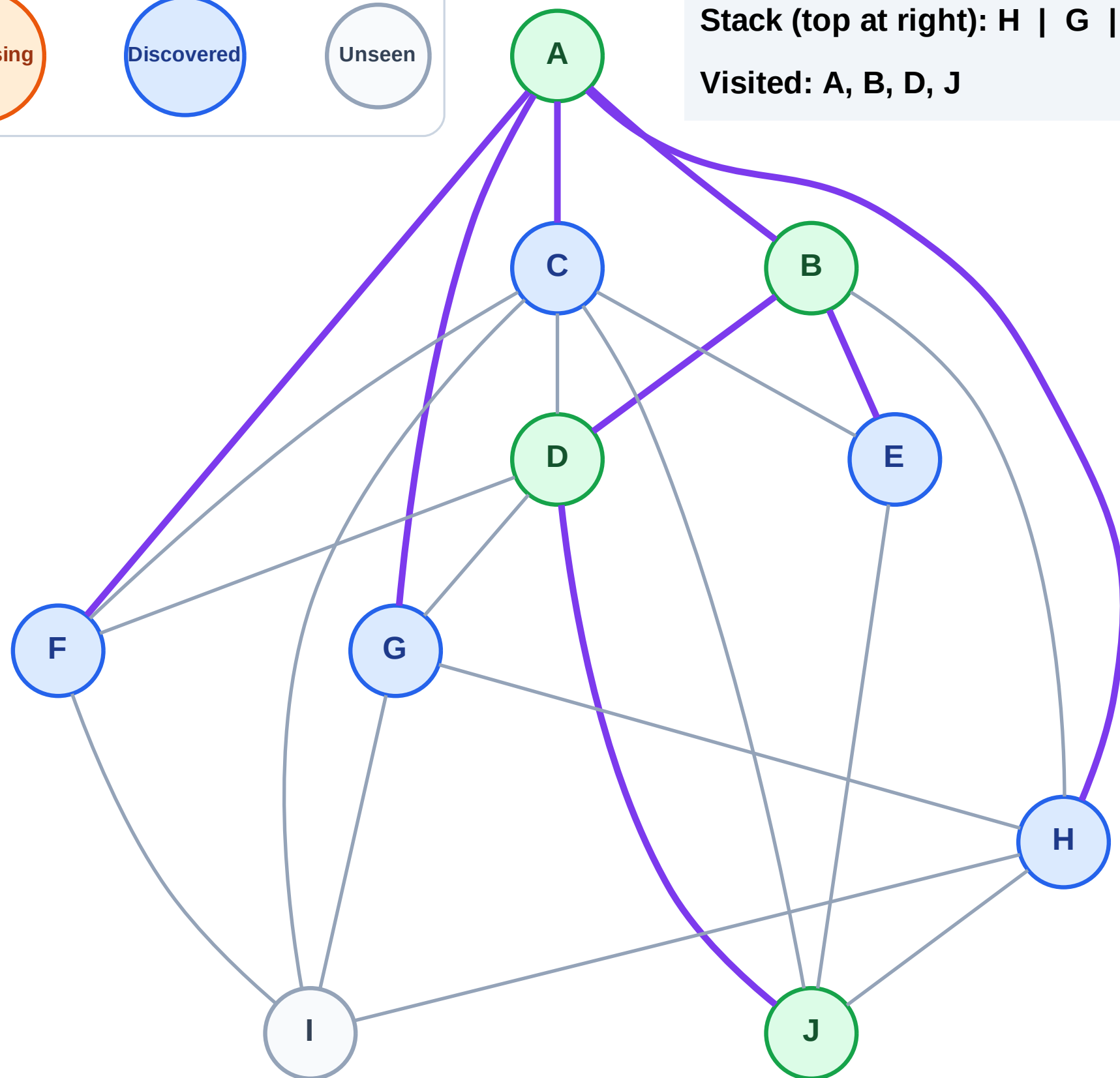
Step 9: No new neighbors from J

## Legend



Stack (top at right): H | G | F | C | E

Visited: A, B, D, J



# DFS Traversal

Step 10: Process node E

Legend

Complete

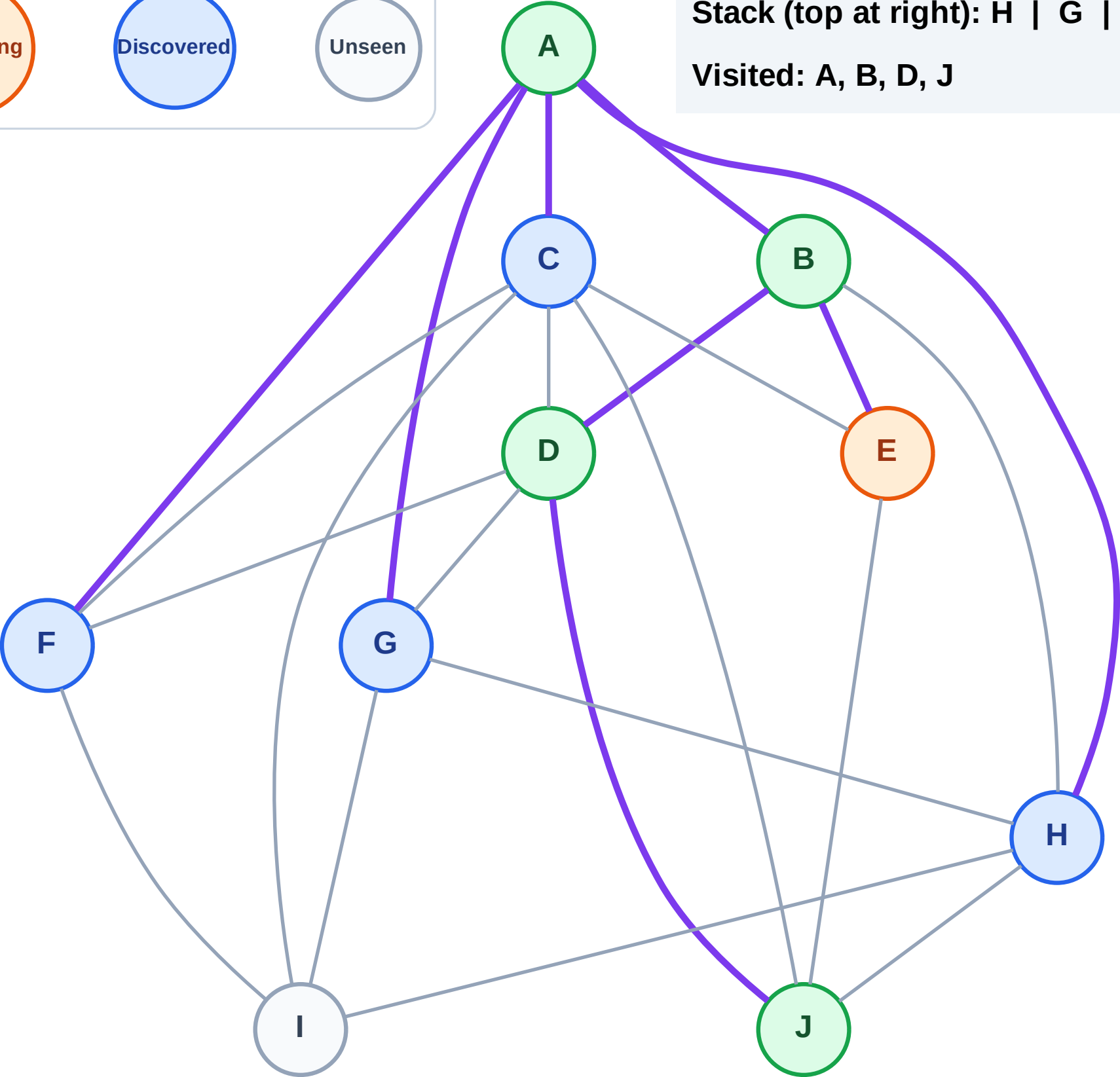
Processing

Discovered

Unseen

Stack (top at right): H | G | F | C

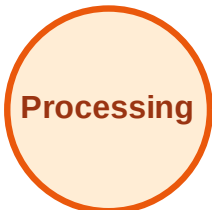
Visited: A, B, D, J



# DFS Traversal

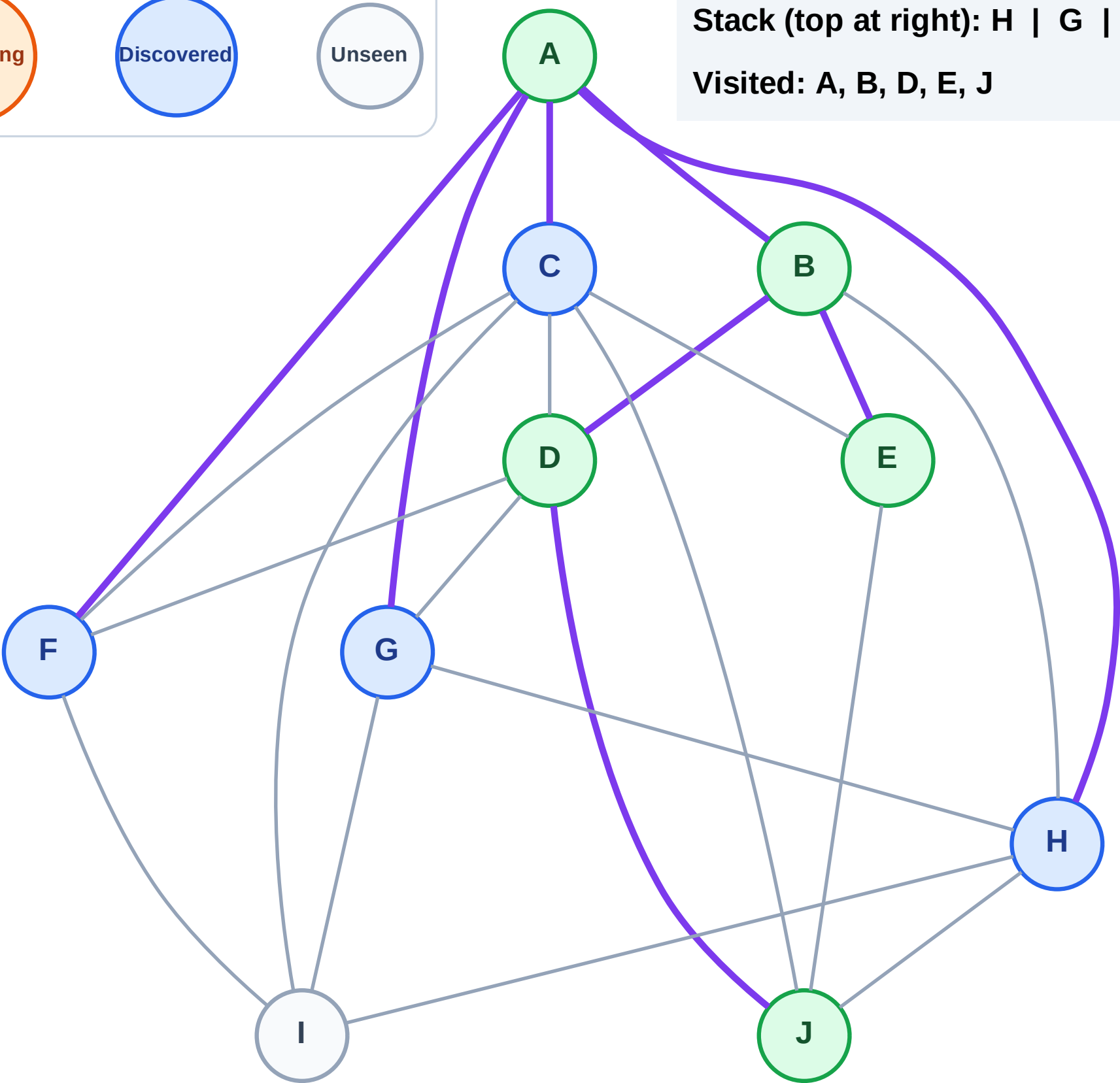
Step 11: No new neighbors from E

## Legend



Stack (top at right): H | G | F | C

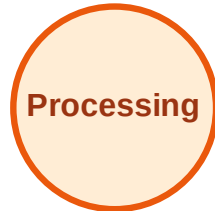
Visited: A, B, D, E, J



# DFS Traversal

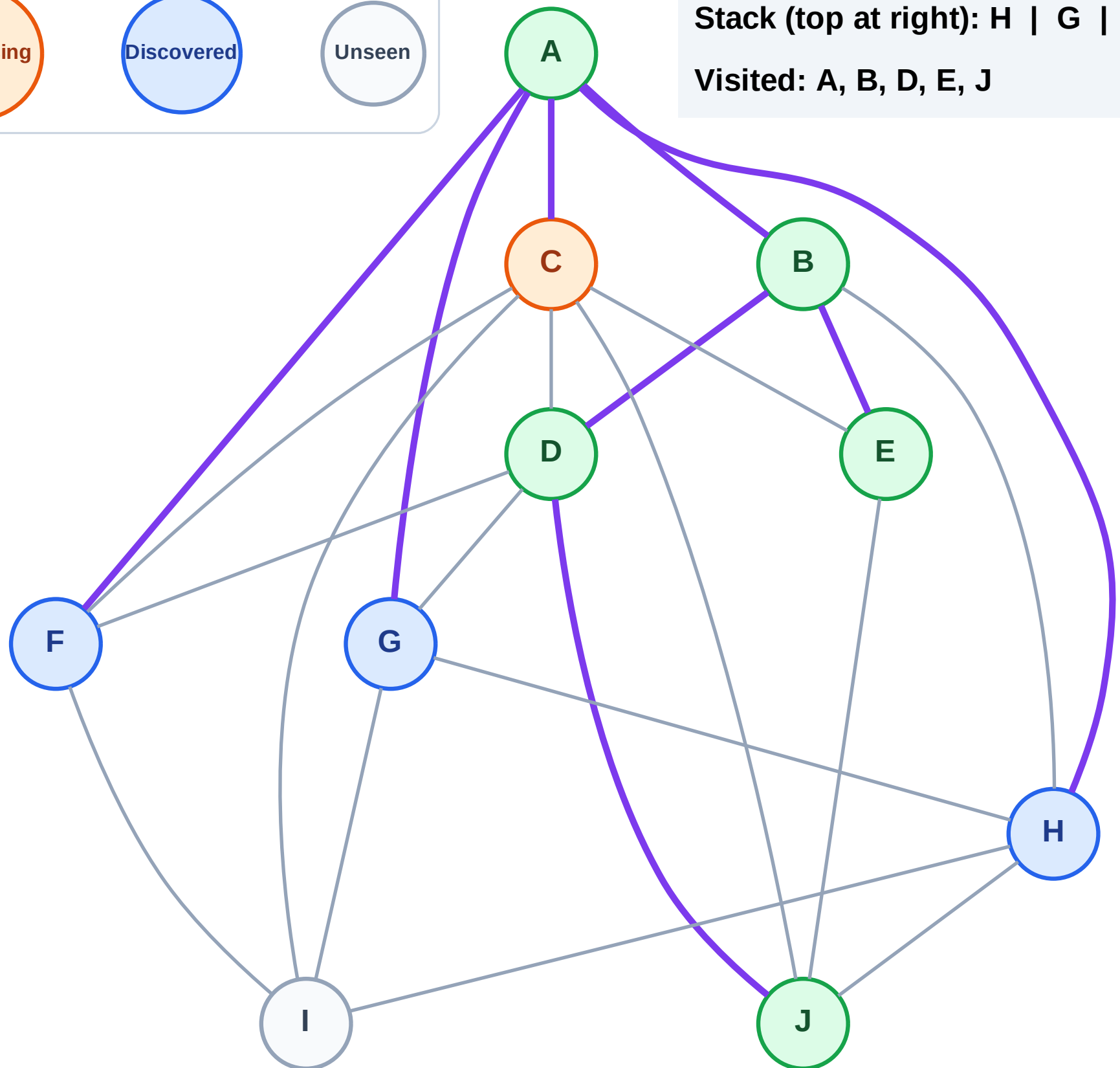
Step 12: Process node C

## Legend



Stack (top at right): H | G | F

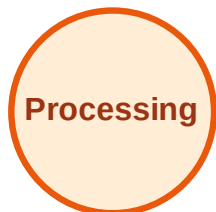
Visited: A, B, D, E, J



# DFS Traversal

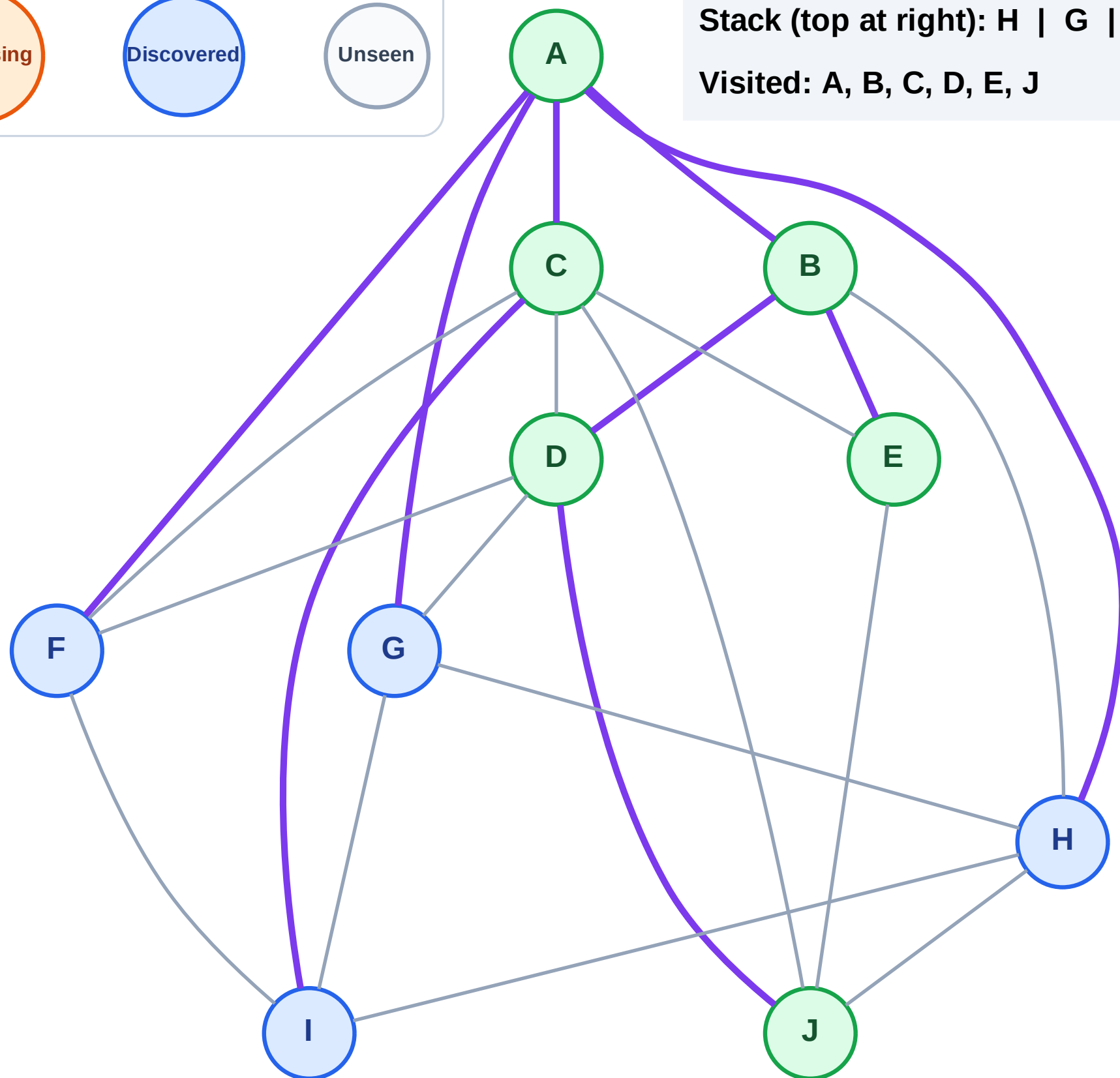
Step 13: Discover I from C

## Legend



Stack (top at right): H | G | F | I

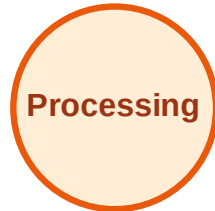
Visited: A, B, C, D, E, J



# DFS Traversal

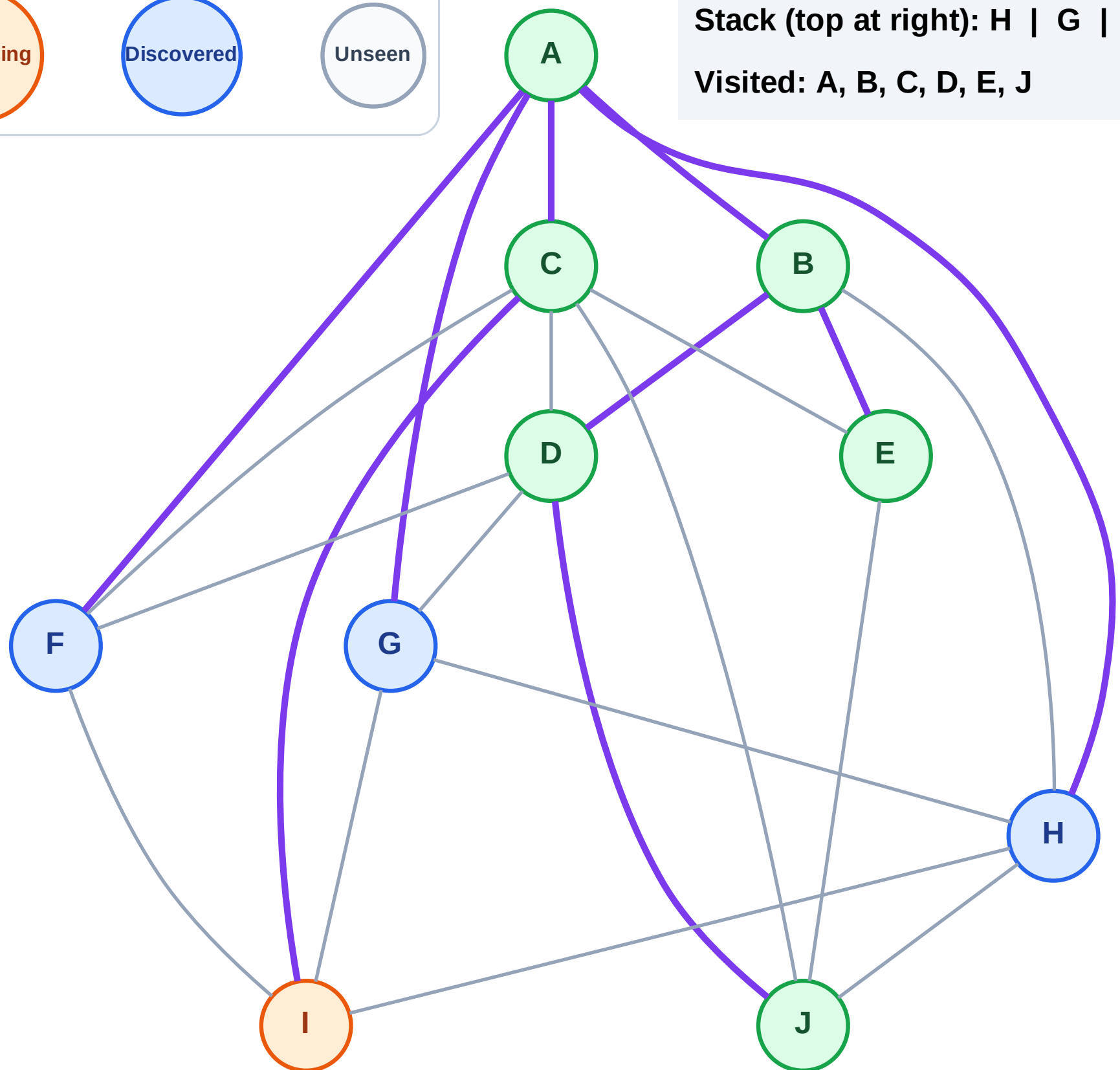
Step 14: Process node I

## Legend



Stack (top at right): H | G | F

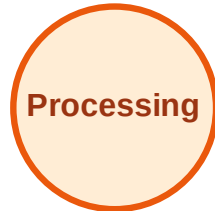
Visited: A, B, C, D, E, J



# DFS Traversal

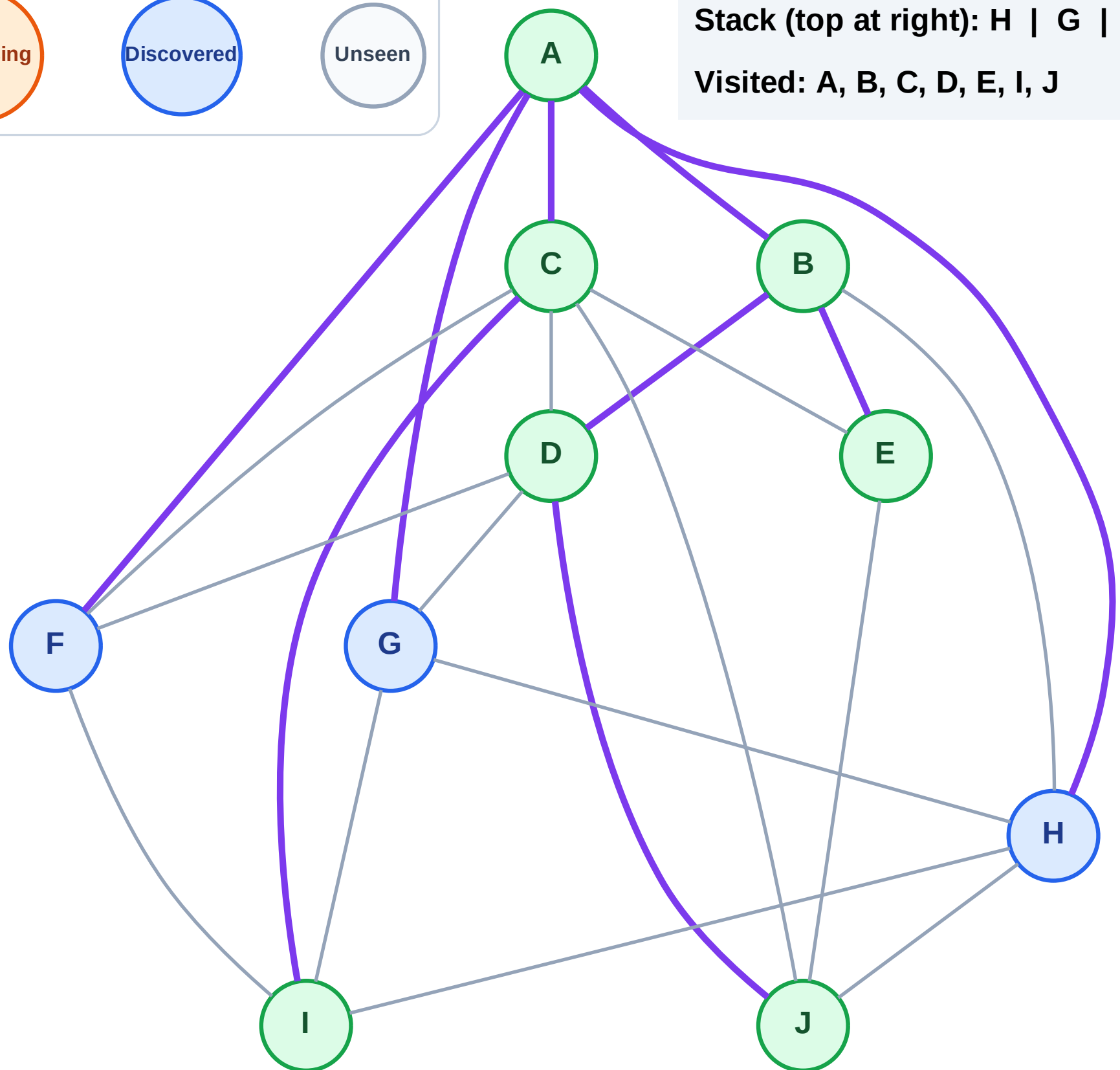
Step 15: No new neighbors from I

## Legend



Stack (top at right): H | G | F

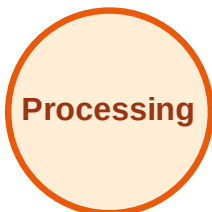
Visited: A, B, C, D, E, I, J



# DFS Traversal

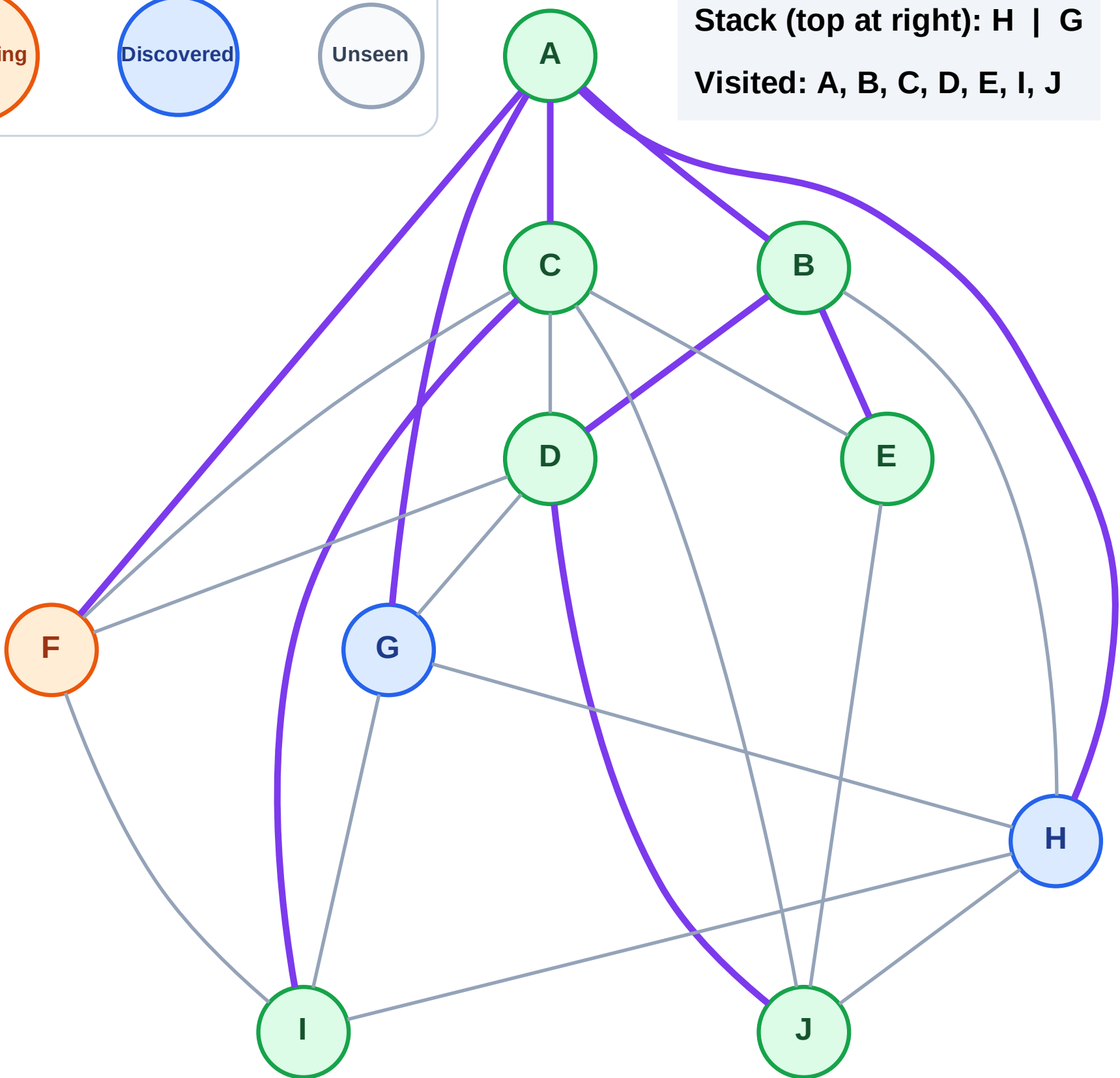
Step 16: Process node F

## Legend



Stack (top at right): H | G

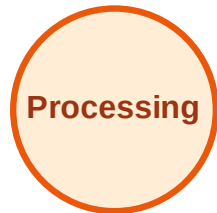
Visited: A, B, C, D, E, I, J



# DFS Traversal

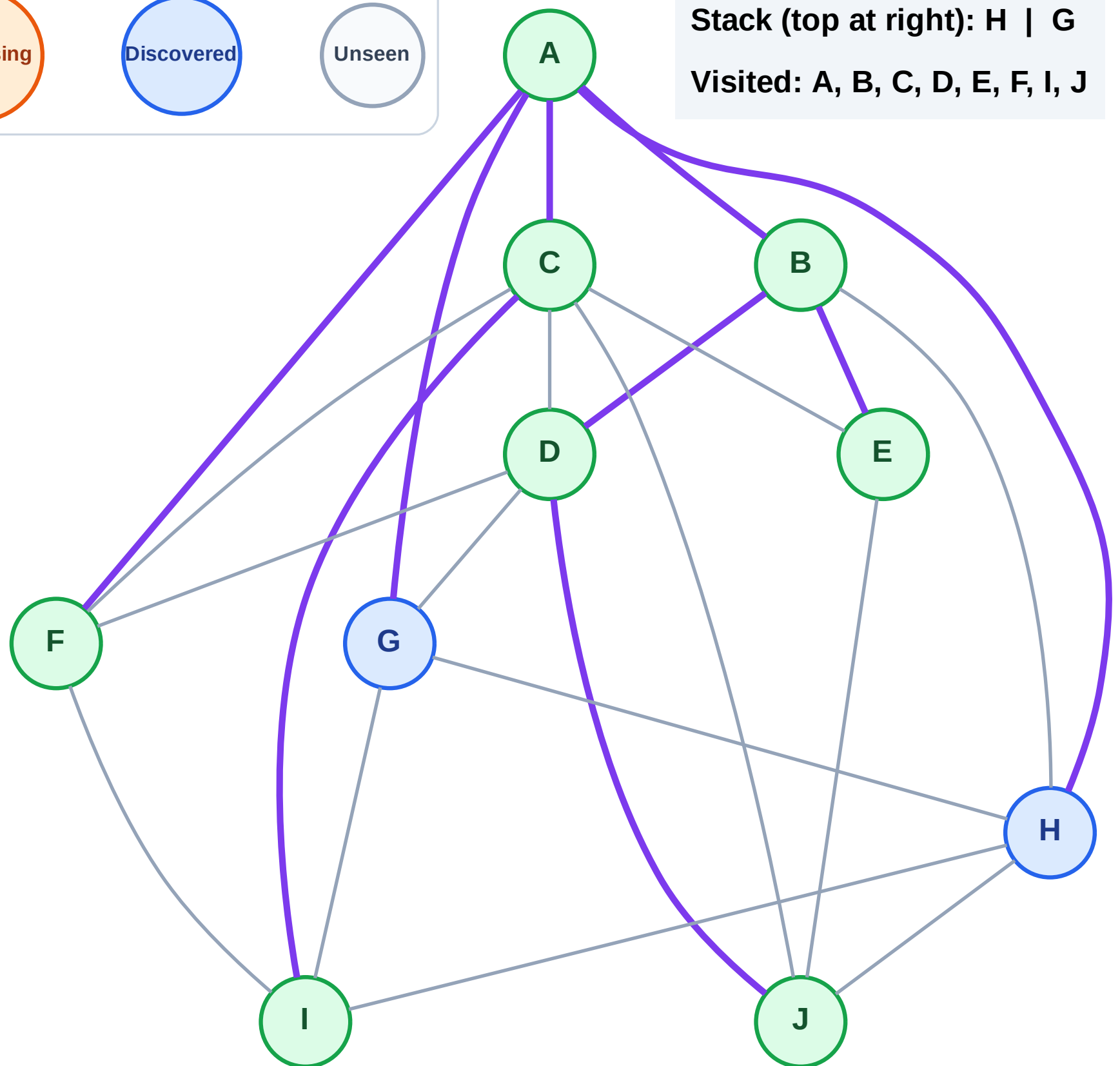
Step 17: No new neighbors from F

## Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



# DFS Traversal

Step 18: Process node G

## Legend

Complete

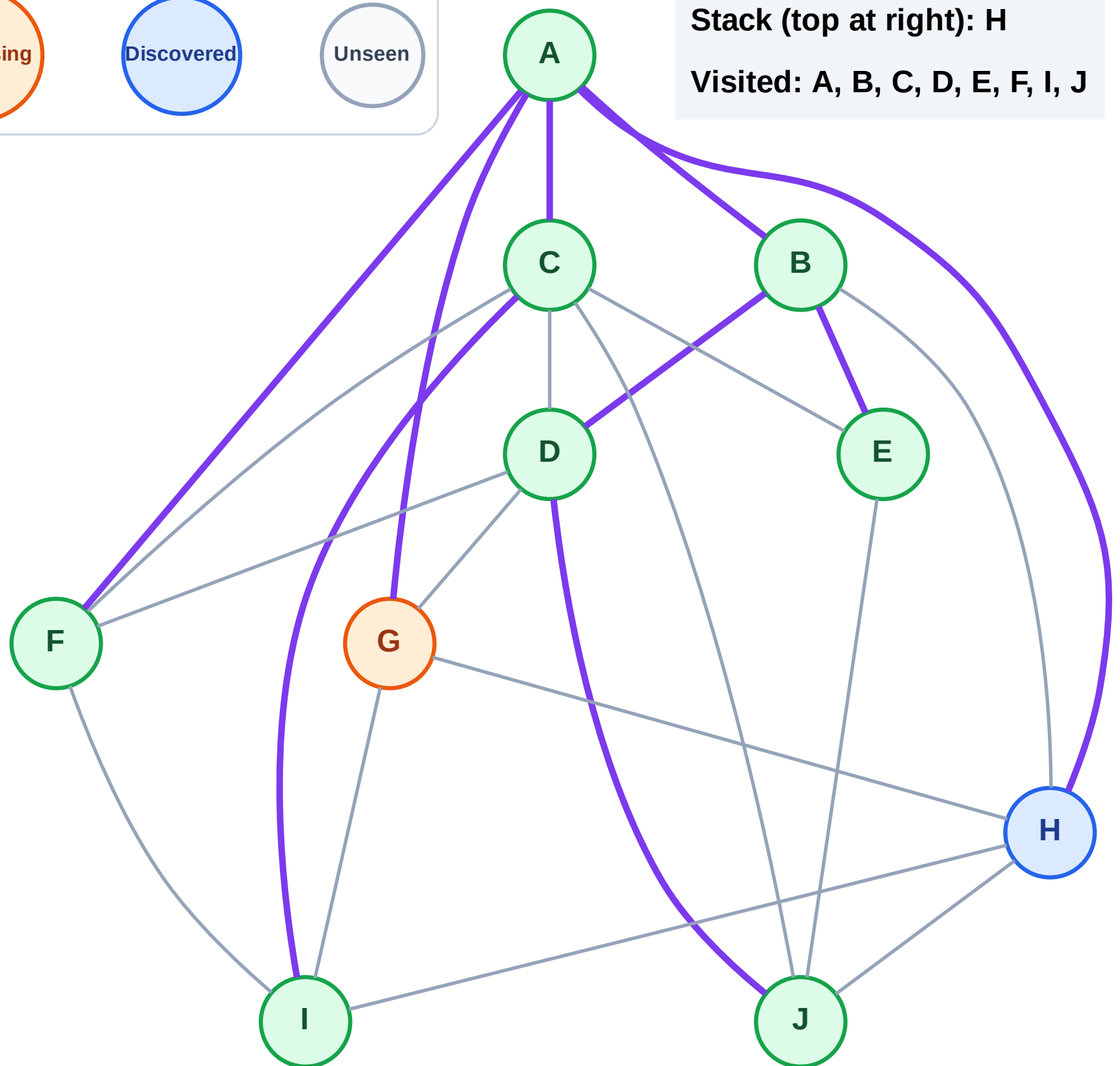
Processing

Discovered

Unseen

Stack (top at right): H

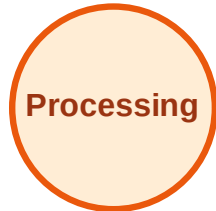
Visited: A, B, C, D, E, F, I, J



# DFS Traversal

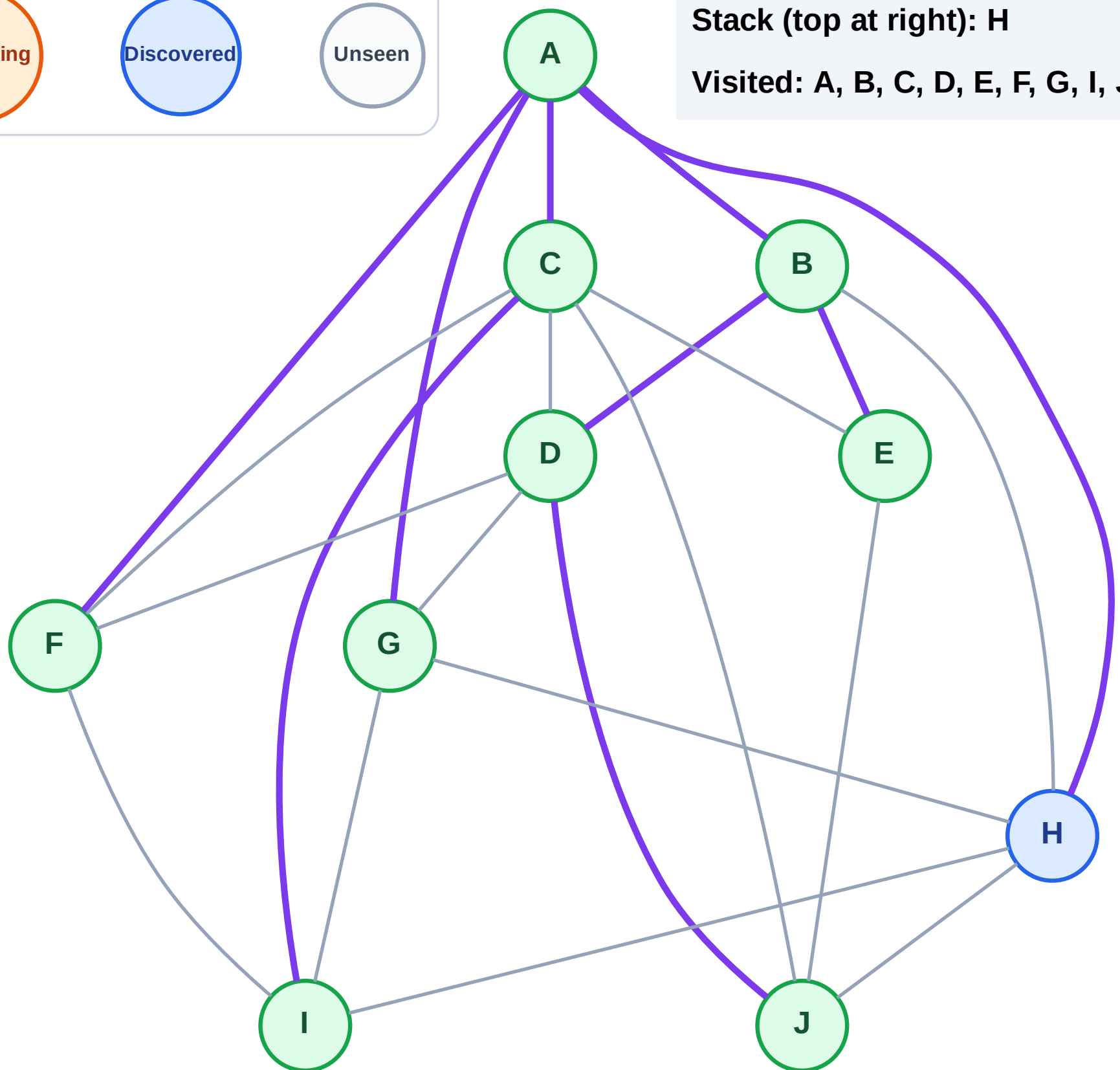
Step 19: No new neighbors from G

## Legend



Stack (top at right): H

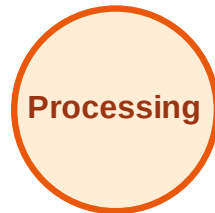
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

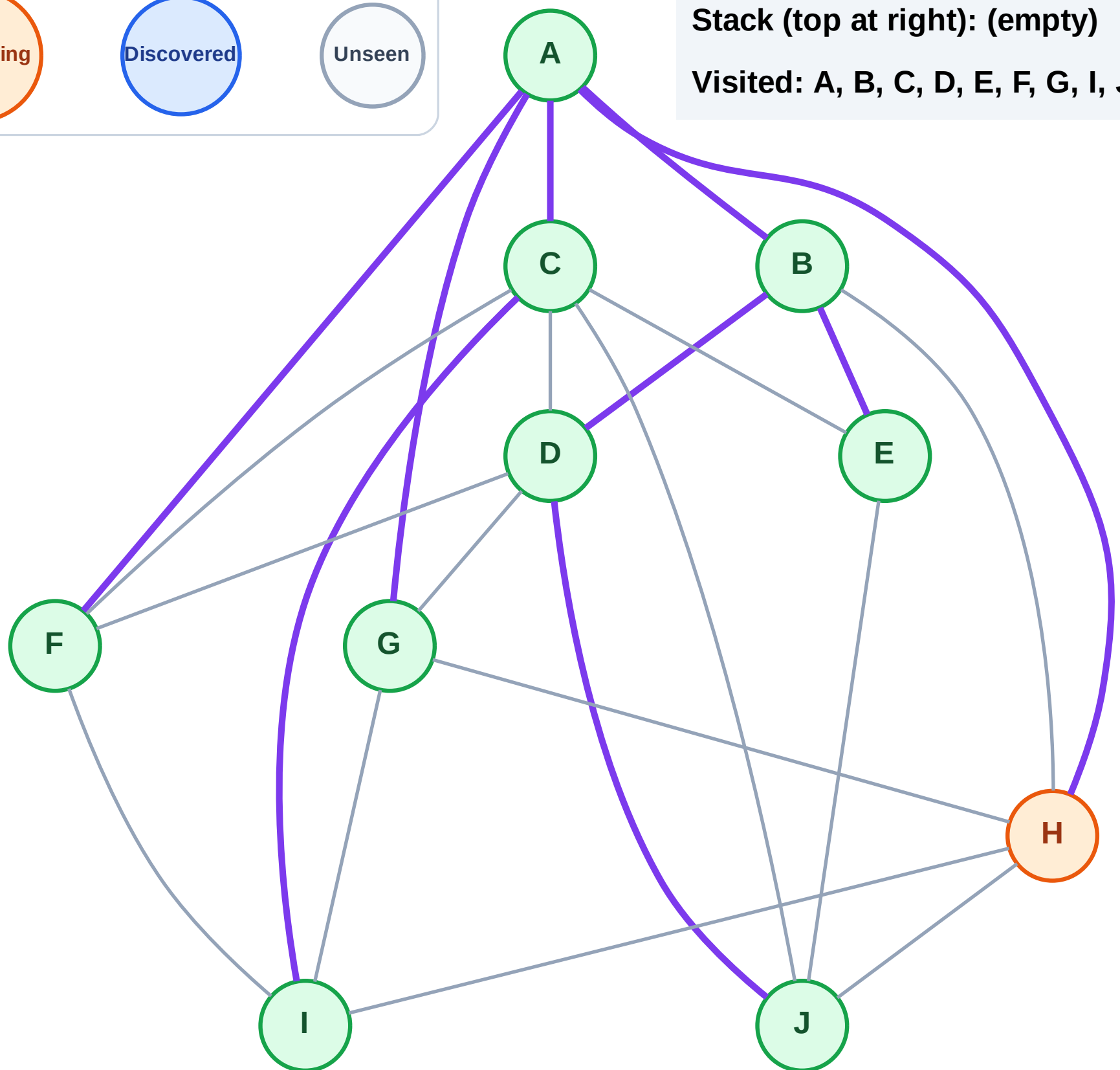
Step 20: Process node H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

Step 21: No new neighbors from H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

